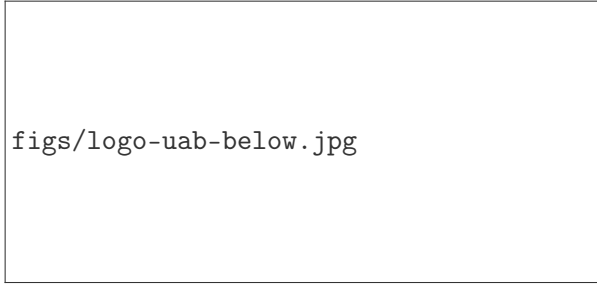


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Nomenclature

Greek Symbols

π $\simeq 3.14\dots$

Acronyms / Abbreviations

AMO Atlantic Multidecadal Oscillation

BSISO Boreal Summer Intraseasonal Oscillation

ENSO El Niño–Southern Oscillation

MJO Madden-Julian Oscillation

NAO North Atlantic Oscillation

PC Principal Component

PDO Pacific Decadal Oscillation

VC Vector of Coefficients

Part I

Embedding

Introduction

- complexity of our environment: more and more data are available (Earth observation data, satellites, reanalysis, citizen science) to describe very complex systems; research becomes more and more data-driven
- dimensionality reduction methods and non-linear modelling approaches are important to get a better understanding
- software tools are necessary to lower the barrier to entry for non-experts

Theoretical Foundations

2.1 Principal Component Analysis

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of a dataset while preserving most of the variance. This is achieved by identifying the principal axes, which are the directions in which the data varies the most. These directions correspond to the eigenvectors of the covariance matrix of the data, with the associated eigenvalues representing the magnitude of variance along these directions. PCA is particularly useful for dimensionality reduction, data compression, feature extraction and visualization of high-dimensional datasets.

The origins of PCA can be traced back to Pearson [128] in 1901 and Hotelling [74] in 1936. It gained significant traction in meteorology through the work of Lorenz [108] in 1956, who introduced the technique under the term *empirical orthogonal function* (EOF) analysis¹. The method has since been extensively applied in meteorology and climate science to identify dominant modes of variability in the atmosphere and oceans, as discussed in numerous textbooks [e.g. 37, 63, 85, 134, 158, 180].

In this section, we derive PCA from the maximum variance formulation², as outlined by Bishop [10, pp. 561-563]. This approach involves projecting the data onto a lower-dimensional linear space, known as the principal subspace, to maximize the variance of the projected data. We will then explore the limitations of PCA, particularly in the context of climate science, and discuss common strategies to address these limitations.

2.1.1 Derivation from Maximum Variance Formulation

Consider a data set represented as a complex-valued, two-dimensional $N \times D$ matrix $\mathbf{Z}$, where N denotes the number of observations (or samples) and D the number of variables (or features).

¹Although EOF analysis remains the preferred term in meteorology and climate science, we will use the more commonly recognized term, PCA.

²Hotelling [74] originally proposed this formulation. An alternative derivation, known as the minimum error formulation by Pearson [128], seeks a linear projection that minimizes the average projection error, defined as the mean squared distance between the original data points and their projections. This alternative approach shows that PCA is an optimal solution for approximating data in the squared error sense.

We will refer to the n^{th} observation as the D -dimensional vector $\mathbf{z}_n$. For simplicity, we assume the data is centered, meaning the mean of each variable is zero, i.e. $\bar{\mathbf{z}} = \frac{1}{N} \sum_{n=1}^N \mathbf{z}_n = \mathbf{0}$. The goal is to project the data onto an M -dimensional space ($M < D$) of M latent variables while maximizing the variance of the projected data.

To start, consider a one-dimensional projection ($M = 1$). We define the direction of this space using a D -dimensional unit³ vector $\mathbf{v}_1$, such that $\mathbf{v}_1^H \mathbf{v}_1 = 1$, where $\mathbf{v}_1^H$ denotes the conjugate transpose of $\mathbf{v}_1$. The data points $\mathbf{z}_n$ are then projected onto this direction, $\mathbf{v}_1^H \mathbf{z}_n$, and the variance of the projected data is

$$\frac{1}{N-1} \sum_{n=1}^{N-1} (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}}) (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}})^H = \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1, \quad (2.1)$$

where $\mathbf{S}_{ZZ}$ is the covariance matrix of $\mathbf{Z}$ given by

$$\mathbf{S}_{ZZ} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{z}_n \mathbf{z}_n^H = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z}. \quad (2.2)$$

The goal is to find the vector $\mathbf{v}_1$ that maimizes this variance, formulated as:

$$\underset{\mathbf{v}_1}{\operatorname{argmax}} \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 \quad \text{subject to} \quad \mathbf{v}_1^H \mathbf{v}_1 = 1. \quad (2.3)$$

To solve this optimization problem, we maximize the projected variance $\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. To enforce the normalization constraint $\mathbf{v}_1^H \mathbf{v}_1 = 1$, a Lagrange multiplier λ_1 is introduced, leading to the unconstrained maximization of:

$$\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^H \mathbf{v}_1). \quad (2.4)$$

By setting the derivative with respect to $\mathbf{v}_1$ to zero, we find a stationary point when:

$$\mathbf{S}_{ZZ} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1. \quad (2.5)$$

³This normalization is for convenience and not necessary, since we are only interested in the direction of $\mathbf{v}_1$, not its magnitude.

This indicates that $\mathbf{v}_1$ must be a eigenvector of $\mathbf{S}_{ZZ}$, with λ_1 being the corresponding real-valued⁴ eigenvalue. The variance $\mathbf{v}_1^T \mathbf{S}_{ZZ} \mathbf{v}_1$ reaches its maximum when $\mathbf{v}_1$ aligns with the eigenvector associated with the largest eigenvalue⁵ λ_1 . The eigenvector is referred to as the first **vector of coefficients**⁶ (VC) of the data set.

To find additional VCs, we iteratively select new directions that maximize the projected variance while being orthogonal to the previously chosen directions. For an M -dimensional projection space, the optimal linear projection that maximizes the variance of the projected data is defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the covariance matrix $\mathbf{S}_{ZZ}$, corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This result can be derived using proof by induction, leading to the concise expression:

$$\mathbf{S}_{ZZ} \mathbf{V} = \mathbf{V} \mathbf{\Lambda}, \quad (2.6)$$

where $\mathbf{V}$ is unitary, i.e. $\mathbf{V}^T \mathbf{V} = \mathbf{I}_M$.

Thus, the eigenvectors of $\mathbf{S}_{ZZ}$ emerge as the new coordiante system "along" regularities in the data. For cases where these regularities are not aligned in unitary directions, see Section 2.1.2.

According to the spectral theorem for normal matrices [e.g. 73, p. 133], the hermitian covariance matrix $\mathbf{S}_{ZZ}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding real-valued eigenvalues λ_i :

$$\mathbf{S}_{ZZ} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^H. \quad (2.7)$$

Each eigenvalue λ_i represents the amount of variance captured by the corresponding component, typically expressed as a fraction γ_i of the total variance:

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (2.8)$$

⁴This follows from the property of the covariance matrix being hermitian.

⁵This can easily verified by left-multiplying Equation (2.5) with $\mathbf{v}_1^H$ and making use of the normalization condition.

⁶Naming conventions can vary significantly across different scientific domains. For example, in machine learning, the terms principal components and principal component scores refer to the eigenvectors (here $\mathbf{V}$) and their corresponding data projections (here $\mathbf{P}$) [e.g. 10]. In statistics, these are often called vectors of coefficients/loadings and principal components [e.g. 85], while in climate science and meteorology, they are known as empirical orthogonal functions and principal components/expansion coefficients [e.g. 158]. In this thesis, I aim to maintain consistency with the terminology commonly used in statistics.

Projecting the data $\mathbf{Z}$ onto the M -dimensional subspace thus provides a lower dimensional representation, called the principal components (PCs),

$$\mathbf{P} = \mathbf{Z}\mathbf{V}, \quad (2.9)$$

which also enables the reconstruction of the original data from the projected data:

$$\mathbf{Z} = \mathbf{P}\mathbf{V}^H. \quad (2.10)$$

However, calculating the eigenvector decomposition is not always computationally efficient. A more efficient method involves using Singular Value Domposition (SVD) [e.g. 73, p. 150], which decomposes the data matrix as

$$\mathbf{Z} = \mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H, \quad (2.11)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are unitary matrices, $\mathbf{U}'^H\mathbf{U}' = \mathbf{V}'^H\mathbf{V}' = \mathbf{I}_M$, and $\mathbf{\Sigma}$ is a diagonal matrix containing the singular values σ_i of $\mathbf{Z}$. The covariance matrix $\mathbf{S}_{ZZ}$ can then be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1}\mathbf{Z}^H\mathbf{Z} = \frac{1}{N-1}\mathbf{V}'\mathbf{\Sigma}^H\mathbf{U}'^H\mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^H, \quad (2.12)$$

where, by comparison with (2.6), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1}\mathbf{\Sigma}^T\mathbf{\Sigma}$. This establishes a direct relationship between the eigenvalues of the covariance matrix and the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (2.13)$$

Comparing (2.10) with (2.11) shows that the PC scores $\mathbf{P}$ can be normalized

$$\mathbf{U} \equiv \mathbf{U}' = \mathbf{P}\mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}}\mathbf{P}\mathbf{\Lambda}^{-1/2},$$

or alternatively $\mathbf{P} = (N-1)^{1/2}\mathbf{U}\mathbf{\Lambda}^{1/2}.$ (2.14)

Thus, $\mathbf{U}^H \mathbf{U} = \mathbf{I}_M$ ensures that the PC scores $\mathbf{U}$ are unitary and uncorrelated.

A common scaling method for the vector of coefficients $\mathbf{V}$ involves scaling by the square root of the corresponding eigenvalues, referred to as **loadings**:

$$\tilde{\mathbf{V}} = \mathbf{V} \mathbf{\Lambda}^{1/2}. \quad (2.15)$$

Although not strictly unitary, these loadings form a generalized unitary matrix, as shown by

$$\tilde{\mathbf{V}}^H \tilde{\mathbf{V}} = \mathbf{\Lambda}. \quad (2.16)$$

Using this scaling, the Equation 2.10 can be expressed as

$$\mathbf{X} = (\mathbf{N} - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^T. \quad (2.17)$$

Computational Considerations The computational cost of performing eigendecomposition on the covariance matrix scales with $O(D^3)$, making it impractical for modern datasets, such as those in climate science, where typical data sizes are on the order of $D = O(10^6)$ or even larger. For example, ERA5 [31], a widely used reference for state-of-the-art reanalysis climate data, distributes a single climate variable on a regular 0.25° grid, resulting in $D = 1\,036\,800$ grid cells. In practice, more efficient algorithms, such as SVD or power methods [55], have been employed traditionally. Recently, randomized low-rank approximation techniques [38, 60] have further reduced computational complexity to $O(ND \log(M) + (N + D)M^2)$. These advancements offer feasible alternatives for handling large, out-of-memory datasets (see Table 2.1).

Shortcomings Despite its widespread use and versatility, PCA has several limitations and potential drawbacks that should be considered:

1. **Linearity Assumption:** PCA assumes a linear structure in the data, which may not hold in complex systems such as climate data, where nonlinear relationships can be significant [90, 152].
2. **Interpretability of VCs:** The orthogonality of the vectors of coefficients is a mathematical construct that may not always correspond to the physical reality of the data, potentially leading to misinterpretations [141].

3. **Variance Focus:** Focusing on directions of maximum variance may not always capture the most relevant features for a specific application, potentially critical but less variable information [46].
4. **Static Nature:** As a static method, PCA does not account for temporal dynamics or changes over time, which is particularly limiting in climate science, where time evolution is a key aspect.
5. **Sensitivity to Outliers:** PCA can be sensitive to outliers, as extreme values can disproportionately influence the VCs, leading to misleading results [22].
6. **Sensitivity to Scaling:** The results are sensitive to the scaling of the input data, meaning that variables with larger ranges can dominate the VCs unless the data is appropriately standardized.
7. **Loss of Information:** While dimensionality reduction is often the goal, PCA inherently discards some information, which could lead to the loss of important details crucial for accurate analysis and interpretation in specific applications.

Over the past few decades, numerous modifications and extensions have been proposed to address these limitations, especially in the context of climate science. For instance, to address the linearity assumption, various forms of kernel PCA [17, 152, 176] have been developed. For improved interpretability, rotated [141] and sparse [86] PCA methods have been proposed. Robust PCA [22] offers a more resilient approach to handling sensitivity to outlier. Furthermore, extended [15, 175] and complex [7, 71] PCA methods have been introduced to account for temporal dynamics. In addition, eigenmethods like Optimal Persistence Analysis (OPA) [35, 36] and Principal Oscillation Pattern (POP) analysis [68] provide alternative approaches independent to variance maximization to understanding dominant patterns in physical climate fields.

In the following sections, we will explore some of these shortcomings and proposed solutions in greater detail. Specifically, we will discuss how additional rotation of the PCA solution can enhance physical interpretability (Section 2.1.2) and how complexifying the data can provide insights into time evolution patterns (Section 2.1.3).

Weighting of Data Variables In some cases, it may be necessary to weight the data variables differently. This can be achieved by scaling the data matrix $\mathbf{Z}$ by a diagonal matrix $\mathbf{W}$, where the j^{th} diagonal element w_j corresponds to the weight of the j^{th} variable. A very common situation is that the observed variables exhibit stark differences in their variance, which can sometimes lead to mask structure in the data as the PCA solution is solely driven by the different scales of variance (so the first principal axis will closely align to the variable with the highest variance) in which case the obvious choice is to use the inverse of the standard

Table 2.1 — Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [10] and Halko *et al.* [60].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

deviation of each variable, $w_j = 1/\sigma_j$, which ensures that all variables are equally weighted in which case the covariance matrix becomes the correlation matrix. Another common situation in climate science is that the physical fields are represented on regular longitude-latitude grids. This representation leads to a higher density of grid points near the poles, which typically results in an exaggerated importance of components spatially focused near the poles. To address this, the data matrix can be weighted by the square root of the cosine of the latitude [123], $w_j = \cos(\text{lat}_j)$, where lat_j is the latitude of the j^{th} grid point. This weighting ensures that the grid points are equally weighted in the PCA solution, regardless of their geographical location. In the following we will discuss some popular extensions of PCA together with a related multi-set method called Maximum Covariance Analysis (see Section 2.2), for which the latitude weighting can be applied in the same way.

2.1.2 Sparse PCA via Rotation

PCA is widely used for dimensionality reduction, as it maximizes the explained variance and provides orthogonality, making it optimal in a least-squares sense [10]. It effectively identifies principal axes that capture the covariance among variables, allowing for the examination of interrelationships between stations, grid points or variables. However, PCA is not ideal for identifying the cause of variation or deriving physical modes of variability [85, 100, 141].

One limitation is that real-world processes do not necessarily adhere to the orthogonality imposed by PCA's eigenvectors. Additionally, the dense nature of these VCs – where all elements are non-zero – can make interpretation challenging. In the context of climate and environmental science, where variables are often represented across spatial grids, these limitations can lead to recognizable but artificial patterns known as Buell patterns [16, 99, 141]. These patterns typically start with a positive component and evolve into increasingly complex structures, such as bipole or tripole patterns (Figure 2.1). Such issues highlight the disconnect between standard PCA and underlying physical processes [141], prompting the development of alternative methods that produce sparse VCs, characterized by many zero or near-zero elements. This sparsity – which necessarily implies the loss of orthogonality for at least one of

the VCs and PC scores – can enhance interpretability by localizing the patterns and reducing the density of the VCs.

In climate science, the most common approach to achieving sparse PCA is through rotation techniques [72, 140, 141], a concept borrowed from factor analysis [cf. 85, pp. 269]. In this "factor analysis viewpoint" of PCA [85, pp. 150-161], a linear transformation is applied to a subset of the VCs to derive a new set of rotated VCs that are "simpler" in a heuristically defined manner, ideally aligning better with the underlying physical processes.

As noted in Equation (2.17), once the orthogonality constraint is relaxed, PCA solutions exhibit inherent indeterminacy, allowing any non-singular transformation $\mathbf{R}$ to satisfy the PCA formulation [e.g. 141, p. 305]:

$$\mathbf{Z} = (\mathbf{N} - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^H = (\mathbf{N} - 1)^{1/2} \mathbf{U} \mathbf{R} \mathbf{R}^{-1} \tilde{\mathbf{V}}^H. \quad (2.18)$$

Rotated PCA exploits this indeterminacy by choosing $\mathbf{R}$ to optimize a simplicity criterion f , resulting in rotated loadings⁷ $\tilde{\mathbf{V}}_R = \tilde{\mathbf{V}} \mathbf{R}$ that maximize f :

$$\underset{\mathbf{R}}{\operatorname{argmax}} f(\tilde{\mathbf{V}}_R). \quad (2.19)$$

There are two main categories of rotation: *orthogonal*⁸ and *oblique*. In orthogonal rotation, $\mathbf{R}$ is a unitary matrix, preserving the uncorrelated nature of the PCs since $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{U}^H \mathbf{U} \mathbf{R} = \mathbf{I}_M$. In oblique rotation, $\mathbf{R}$ is a general non-singular matrix, resulting in correlated PCs, as $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{R}$. Regardless of the method, the rotated loadings are not orthogonal,

$$\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R = \mathbf{R}^H \tilde{\mathbf{V}}^H \tilde{\mathbf{V}} \mathbf{R} \stackrel{(2.16)}{=} \mathbf{R}^H \mathbf{A} \mathbf{R} \neq \mathbf{I}_M, \quad (2.20)$$

which is crucial for achieving a more interpretable solution. Various simplicity criteria f have been proposed [see e.g. 141], with most focusing on making the rotated loadings more localized

⁷In principle, it is possible to rotate the unitary matrix of coefficients $\mathbf{V}$ instead of the loadings matrix $\tilde{\mathbf{V}}$ [85, 115]. However, this approach is seldom used in practice. The primary reason is that rotating the unitary coefficient matrix preserves the orthogonality of the VCs but results in the loss of orthogonality among the PCs, which does not address issues like the Buell patterns. Moreover, this method does not consider the relative importance of the spatial patterns, as indicated by the eigenvalues, making the results of the rotation process highly sensitive to the selection of the number of components to rotate.

⁸In the broader context of complex data, this would be referred to as a unitary rotation. However, since factor analysis has primarily been applied to real-valued data, the terms orthogonal and oblique rotations have become the standard. To avoid confusion, I use these historical terms while acknowledging that they also encompass the more general case of complex data.

and physically interpretable. In practice, the specific choice of simplicity criterion often has little impact on the results [141].

Varimax Orthogonal Rotation The most widely used orthogonal rotation technique is the varimax criterion [87], which aims to simplify the VCs by maximizing the variance of the squared loadings,

$$f(\tilde{\mathbf{V}}_R) = \sum_{m=1}^M \left(p \sum_{j=1}^P b_{jm}^2 - \left(\sum_{j=1}^P b_{jm} \right)^2 \right) \quad (2.21)$$

where b_{jm} are the squared norm of the rotated loadings $|\tilde{v}_{R,jm}|^2/h_j^2$ normalized by the communalities $h_j^2 = \sum_{m=1}^M |\tilde{v}_{R,jm}|^2$ of variable j . The varimax rotation pushes the elements of $\tilde{\mathbf{V}}_R$ towards either 0 or 1, creating a sparse solution with primarily large or small values, which is particularly useful for identifying localized physical processes, such as atmospheric or oceanic teleconnections.

Promax Oblique Rotation The promax criterion [69], a generalization of the varimax criterion, is one of the most common oblique rotation techniques. It is based on the premise that an orthogonal simple structure solution, such as varimax, is often close to an optimal oblique solution. The promax method constructs a hypothetical oblique target matrix $\mathbf{T}$ by performing an oblique Procrustes transformation [81] on the varimax-rotated loadings matrix,

$$\mathbf{T} = \tilde{\mathbf{V}}_R \mathbf{R} + \mathbf{E}, \quad (2.22)$$

where $\mathbf{R}$ is the transformation matrix and $\mathbf{E}$ is a matrix of residuals. The normal equation for solving the complex Procrustes problem is given by:

$$\mathbf{R} = (\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R)^{-1} \mathbf{T}^H \tilde{\mathbf{V}}_R. \quad (2.23)$$

To define the target matrix $\mathbf{T}$, the promax criterion increases the discrimination between high and low loadings by raising the loadings to a power q while retaining the original sign,

$$t_{jm} = \frac{|a_{jm}|^{q+1}}{a_{jm}}, \quad (2.24)$$

where q controls the degree of obliqueness and a_{jm} are the elements of the max-normalized, varimax-rotated loadings matrix $\mathbf{A}$, defined as

$$\mathbf{A} = \tilde{\mathbf{A}}\mathbf{K}^{-1}, \quad (2.25)$$

$$\tilde{\mathbf{A}} = \mathbf{H}^{-1}\tilde{\mathbf{V}}_R, \quad (2.26)$$

with $\mathbf{K}$ being a diagonal matrix containing the maximum absolute values of each column of $\tilde{\mathbf{A}}$, $a_m = \max_m |A_{pm}|$, and $\mathbf{H}$ a diagonal matrix with the square root of the communalities h_j for the j^{th} variable.

Application and Criticism Both orthogonal and oblique rotation techniques, particularly varimax and promax, have been extensively used since the 1980s to identify stationary modes of variability in climate data [e.g. 8, 51, 72, 82, 98, 106, 116]. Notable climate oscillations, such as the North Atlantic Oscillation (NAO), have been defined based on varimax-rotated patterns of loadings [8]. These techniques also help identify "ghost patterns", artificial patterns identified by PCA that are not physically meaningful [82, 105].

Rotated modes offer several advantages, such as greater robustness against temporal and spatial subsampling [25] compared to unrotated modes. They also effectively mitigate the statistical artifacts associated with Buell patterns, resulting in more realistic representations of the underlying autocorrelation structure and thus increased interpretability [105, 141, 142]. Moreover, rotation techniques have been successfully integrated into other PCA extensions dealing with the non-linear and non-stationary nature of climate oscillations [17, 59, 71].

However, rotation techniques are not without criticism, mainly due to the ad-hoc nature of the various simplicity criteria [cf. 85, pp. 270-274]. This has led to the exploration of alternative approaches to enforce sparsity. Notably, Jolliffe *et al.* [86] introduced the Simplified Component Technique - LASSO (SCoTLASS), a sparse PCA variant based on the least absolute shrinkage and selection operator (LASSO), originally from regularized regression [166]. This method imposes a constraint on the ℓ_1 norm of the vector of loadings, resulting in some loading coefficients being exactly zero.

Despite its advantages, SCoTLASS has drawbacks, particularly when the number of variables D is much larger than the number of observations N . In such cases, SCoTLASS can only select at most N non-zero elements in the loading vectors, which is not ideal in climate science

content/chapter2/figs/example_synthetic_pca_rotated.pdf

Figure 2.1 — Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

where often $D \gg N$. Additionally, since SCoTLASS solutions cannot be obtained via SVD and require a numerical approach using projected gradient methods [168], the computational demand is typically higher than for (rotated) PCA. While this method has inspired many other sparse PCA techniques based on penalties or constraints (usually ℓ_0 or ℓ_1) [see e.g. 44, 185, 186], these formulations are often complex and require algorithmic innovations to avoid issues like local optima and ensure convergence. They may also face challenges with memory limitations, slow computation speeds and limited effectiveness in modelling the data [21, 58].

It is likely for these reasons, that rotated PCA remains one of the most commonly used techniques for achieving sparse PCA solutions in climate science. It is computationally efficient, straightforward to implement and provides a more interpretable understanding of the data compared to standard PCA.

2.1.3 Hilbert PCA

While PCA efficiently identifies the principal axes along which the covariance among all variables is distributed, and rotated PCA provides a more sparse solution related to underlying physical processes, both methods are inherently static. They do not for temporal dynamics or changes over time, as they only consider the current state of the physical field. This limitation is significant in climate science, where time evolution is crucial. For instance, standard PCA does not differentiate between the order of data points over time, resulting in the same eigenvectors and eigenvalues regardless of temporal permutations.

This static nature poses a challenge for capturing oscillatory patterns, such as QBO, MJO, ENSO and BSISO, which involve spatial patterns propagating over time. These phenomena cannot be adequately captured by standard PCA.

Hilbert Transform as a Momentum Operator

To address this limitation, one approach is to model both the current state of the climate and its tendency or 'momentum', analogous to the Hamiltonian principle in mechanics, where a system's future is described by first-order differential equations for location (state) and momentum (tendency) [cf. 158, p. 353]. In classical mechanics, propagating features are represented using canonical coordinates: each state variable is augmented by a momentum variable, allowing oscillations to be depicted as rotations in a two-dimensional phase space. From a data perspective, this augmentation can be achieved via the Hilbert transform ($\mathcal{H}$). For a real time series $x(t)$, the Hilbert transform $x'(t)$ is defined as

$$x'(t) \equiv \mathcal{H}\{x\}(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{\tau - t} d\tau, \quad (2.27)$$

where p.v. denotes the Cauchy principal value ensuring convergence despite the singularity at $\tau = t$. The Hilbert transform emphasises the local properties of $x(t)$ by convolution with $\frac{1}{\pi t}$. In the frequency domain, it is related to the Fourier transform $\mathcal{F}$ by

$$\mathcal{F}\{y\}(\omega) = -i \operatorname{sgn}(\omega) \mathcal{F}\{x\}(\omega) \quad (2.28)$$

where the imaginary unit i indicates a phase shift of $\pi/2$ for the frequency components ω of x , placing x' in quadrature to x . This provides an intuitive interpretation of the Hilbert transform, as it acts as a linear filter that alters the phase of the signal's components by 90 degrees without changing their amplitude.

By using the Hilbert transform, a real-valued time series $x(t)$ can be complexified into $z(t) = x(t) + i\mathcal{H}\{x\}(t)$, known as the *analytic signal*, which allows for the computation of the instantaneous amplitude and phase of $x(t)$:

$$A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x\}(t)^2}, \quad \phi(t) = \arg(z(t)) = \arctan\left(\frac{\mathcal{H}\{x\}(t)}{x(t)}\right). \quad (2.29)$$

Similar results can be obtained for the discrete case [125, pp. 942-956], and this concept can be extended to multivariate data [e.g. 63, pp. 105-106], defining the analytic signal of a data matrix $\mathbf{X}$ as

$$\mathbf{Z} = \mathbf{X} + i\mathcal{H}\{\mathbf{X}\}, \quad (2.30)$$

where $\mathcal{H}\{\mathbf{X}\}$ is the feature-wise Hilbert transform of $\mathbf{X}$.

Rasmusson *et al.* [137] introduced the concept of Hilbert PCA using this data augmentation technique⁹. This method, refined by Barnett [7] and Horel [71], involves applying PCA to the augmented data matrix $\mathbf{Z}$. By incorporating phase information, Hilbert PCA can capture the direction and speed of wave-like patterns, unlike traditional PCA, which only identifies standing oscillations. The resulting eigenvectors and principal component scores are complex, with the real and imaginary components linked via the Hilbert transform. The variance capture by each component is represented by the real eigenvalues of the complex covariance matrix of $\mathbf{Z}$ (see Section 2.1).

The main advantage of Hilbert PCA is the ability to identify and reconstruct propagating waves, which is especially valuable in studying climate phenomena characterized by such patterns, including MJO, ENSO, QBO and BSISO. For instance, Cai & Baines [20] utilized Hilbert PCA to analyze SST in the southern hemisphere, linking the findings to ENSO, while Bürger [19] employed it to develop an index for BSISO. However, to reliably identify oscillatory patterns, sufficiently long time series are required to resolve at least one full oscillation [114]. Additionally, interpreting of the Hilbert transform as representing the instantaneous frequency is valid holds for narrow-band signals; the phase of the Hilbert transform is not well-defined for broadband signals [12, 50]. Moreover, real-time monitoring is challenging with Hilbert PCA because new real-valued data cannot be directly projected onto the existing patterns without knowing or estimating the exact momentum (phase) of the new data, which requires future information (Equation (2.27)). To mitigate this, an approximative real-time (causal) Hilbert transform can be employed [18, 19], offering a practical remedy to the issue of real-time applicability. Finally, Hilbert PCA shares PCA's limitations regarding physical interpretability due to the orthogonality of the VCs. To address this, rotated Hilbert PCA has been proposed to enhance interpretability [11, 72].

⁹Although Rasmusson *et al.* [137] originally termed it *Hilbert Singular Decomposition*, the terms *Hilbert EOF/PCA* and *Complex EOF* analysis are more commonly used in the literature [19, 63, 85, 158].

2.2 Maximum Covariance Analysis

In contrast to finding latent variables that maximize covariance within a single dataset, Maximum Covariance Analysis (MCA) aims to identify the latent variables that are common in two sets of data and maximize covariance. For example, in the case of the MJO [110], which is the dominant mode of intraseasonal variability in the tropical atmosphere, characterized by an eastward-moving cell of overturning wind circulation and associated regions of enhanced and suppressed precipitation, MCA helps in findings shared dynaics between different variables. This method allows us to focus on shared variability while ignoring dominant variations associated with only one variable, such as diurnal cycles linked to surface processes absent at higher atmosphere levels.

MCA identifies patterns of maximum covariance between datasets by finding paris of linear subspaces with high cross-covariance [e.g. 158, p. 321]. Unlike PCA, which operates on a single set of variables, MCA considers two¹⁰ sets of variables and seeks components that exhibit high covariance across datasets. This approach can be formulated as an optimization problem, aiming to find pairs of (complex-valued) coupled patterns $\mathbf{v}_{s,1}$ for datasets $\mathbf{Z}_s$, $s = x, y$, such that the projection maximizes the covariance between the datasets. The solution involves a generalized eigenvector problem, often solved using the Singular Value Decomposition (SVD) of the cross-covariance matrix.

$$\underset{\mathbf{v}_{x,1}, \mathbf{v}_{y,1}}{\operatorname{argmax}} \mathbf{v}_{x,1}^H \mathbf{C}_{xy} \mathbf{v}_{y,1} \quad \text{subject to} \quad \|\mathbf{v}_{x,1}\|_2 = \|\mathbf{v}_{y,1}\|_2 = 1, \quad (2.31)$$

where the cross-covariance matrix between the two datasets is given by

$$\mathbf{C}_{xy} = \frac{1}{N-1} \mathbf{Z}_x^H \mathbf{Z}_y. \quad (2.32)$$

The solution to this problem gives rise to a generalized eigenvalue equations similar to that of PCA, which can also be solved by using the SVD of the cross-covariance matrix

$$\mathbf{C}_{xy} = \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^H \quad (2.33)$$

where the i^{th} column of the left and right singular vectors, $\mathbf{V}_x$ and $\mathbf{V}_y$, contains the pattern vectors $\mathbf{v}_{x,i}$ and $\mathbf{v}_{y,i}$, and $\mathbf{\Sigma}$ a diagonal matrix containing the singular values σ_i in decending

¹⁰In theory, this can be extended to more than two sets of data [88].

order. The first left and right singular vectors correspond to the maximum covariance, given by the largest singular value σ_1 .

Since the pattern vectors are unitary, datasets can be expanded as

$$\mathbf{Z}_s = \mathbf{P}_s \mathbf{V}_s^H, \quad (2.34)$$

with the expansion coefficients $\mathbf{P}_s$, $s \in [X, Y]$. The expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are uncorrelated with each other:

$$\mathbf{P}_x^H \mathbf{P}_y = \mathbf{V}_x^H \mathbf{Z}_x^H \mathbf{Z}_y \mathbf{V}_y \stackrel{(2.32)}{=} (N-1) \mathbf{V}_x^H \mathbf{C}_{xy} \mathbf{V}_y \stackrel{(2.33)}{=} (N-1) \boldsymbol{\Sigma}. \quad (2.35)$$

However, the expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are generally correlated within themselves, i.e. $\mathbf{P}_s^H \mathbf{P}_s$ is not a diagonal matrix.

Next, as in PCA, we are interested in the relative importance of each mode in capturing the shared variability between the two datasets. Note that since MCA maximized the *squared* covariance, it is the total squared covariance that is conserved by the decomposition, which is given by the squared Frobenius norm of the cross-covariance matrix:

$$\|\mathbf{C}\|_F^2 \equiv \left(\sum_i^N \sum_j^P |c_{ij}|^2 \right) = \left(\sum_i^M \sigma_i^2 \right). \quad (2.36)$$

where M is the rank of the cross-covariance matrix. We provide a formal proof for this statement in Appendix A.2.

Therefore, the relative importance of mode k can be quantified by the squared covariance fraction $f_{xy,k}$:

$$f_{xy,k} = \frac{\left\| \sigma_k \mathbf{v}_{x,k} \mathbf{v}_{y,k}^H \right\|_F^2}{\left\| \mathbf{C}_{xy} \right\|_F^2} = \frac{\sigma_k^2}{\sum_{i=1}^M \sigma_i^2}. \quad (2.37)$$

Alternatively, this expression can be formulated as a residual-based measure:

$$f_{xy,k} = 1 - \frac{\|(\mathbf{X} - \mathbf{X}_k)^H(\mathbf{Y} - \mathbf{Y}_k)\|_F^2}{\|\mathbf{X}^H\mathbf{Y}\|_F^2} \quad (2.38)$$

where $\mathbf{X}_k$ and $\mathbf{Y}_k$ represent the reconstructed datasets using the k^{th} mode.

This fraction provides a measure of how much of the shared variability between the two datasets is captured by the k^{th} mode. To assess the importance of the patterns themselves, it is necessary to consider the variance within the individual datasets explained by each mode relative to the total variance of the individual datasets. This can be expressed using the squared Frobenius norm of the individual datasets:

$$f_{ss,k} = \frac{\mathbf{v}_{s,k}^H \mathbf{C}_{ss} \mathbf{v}_{s,k}}{\text{tr}(\mathbf{Z}_s)} = \frac{\mathbf{p}_{s,k}^H \mathbf{p}_{s,k}}{\|\mathbf{Z}_s\|_F^2}. \quad (2.39)$$

Note that this projection-based measure of explained variance is mathematically the same in MCA as the following residual-based measure:

$$f_{ss,i} = 1 - \frac{\|\mathbf{Z}_s - \mathbf{p}_{s,i} \mathbf{v}_{s,i}^H\|_F^2}{\|\mathbf{Z}_s\|_F^2} \quad (2.40)$$

While the squared covariance fraction $f_{xy,i}$ indicates the amount of shared variability between the two datasets, it does not provide information about the strength of the association or coupling between them. To measure this, we use the correlation coefficient ρ_i for the expansion coefficients of mode i between the two datasets:

$$\rho_i = \frac{\text{cov}(\mathbf{p}_{1,i}, \mathbf{p}_{2,i})}{(\text{var}(\mathbf{p}_{1,i}) \text{var}(\mathbf{p}_{2,i}))^{1/2}} = \frac{\sigma_i}{\|\mathbf{p}_{1,i}\|^2 \|\mathbf{p}_{2,i}\|^2}. \quad (2.41)$$

The homogeneous covariance patterns [14] for the pattern vectors are:

$$\mathbf{h}_{M,x,i} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{x,i}) = \mathbf{S}_{xx} \mathbf{v}_{x,i}, \quad (2.42a)$$

$$\mathbf{h}_{M,y,i} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{y,i}) = \mathbf{S}_{yy} \mathbf{v}_{y,i}, \quad (2.42b)$$

These patterns provide an indicator of which variables or features are important for the covarying components in both fields. The corresponding heterogeneous covariance patterns are defined as:

$$\mathbf{h}_{T,x,i} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{y,i}) = \sigma_i \mathbf{v}_{y,i}, \quad (2.43a)$$

$$\mathbf{h}_{T,y,i} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{x,i}) = \sigma_i \mathbf{v}_{x,i}. \quad (2.43b)$$

These patterns indicate how well the variables in one field can be predicted from the i^{th} expansion coefficient of the other field. One may also use standardized data to obtain the respective homogeneous and heterogeneous correlation patterns.

MCA gained popularity in the field of meteorology¹¹ through the works of Bretherton *et al.* [14] and Wallace *et al.* [173], since when it has been extensively applied [e.g. 9, 103, 121, 138, 169, 173].

The method's orthogonal patterns ensure efficient dimensionality reduction of the covariance but limit physical interpretability, similar to PCA. To enhance interpretability, rotation of the MCA solution can be employed, as proposed by Cheng & Dunkerton [24], and later applied by Frankignoul & Kestenare [48]. A 'loaded' composite matrix $\mathbf{L}$ of the left and right singular vectors is constructed and a simplicity criterion (e.g. varimax) is applied,

$$\mathbf{L} = \begin{bmatrix} \mathbf{V}_x \\ \mathbf{V}_y \end{bmatrix} \Sigma^{1/2}, \quad (2.44)$$

where the term 'loaded' refers to the fact that the singular vectors are weighted by the square root of the singular values. The rotated pattern vectors $\mathbf{V}_{R,s}$ are then obtained by applying the rotation matrix $\mathbf{R}$ to the 'loaded' $\mathbf{V}_s$.

$$\mathbf{V}_{R,s} = \mathbf{V}_s \Sigma^{1/2} \mathbf{R} \mathbf{W}_s^{-1} \quad (2.45)$$

where $\mathbf{W}_s^{-1}$ is a diagonal matrix with the elements that normalize the rotated pattern vectors to unit length.

¹¹According to Jolliffe [85, p. 225], the technique was first used in psychology in the 1950s under the name *inter-battery factor analysis*. As a variant of Partial Least Squares (PLS), it was later applied in biomedical sciences by Sampson *et al.* [147] and was referred to as *PLS-SVD* by Wegelin [177]. In meteorology, Bretherton *et al.* [14] introduced the term *SVD analysis*, which remains in use today [e.g. 121]. Occasionally, the method is also called *Canonical Covariance Analysis*, similar to Canonical Correlation Analysis [e.g. 27, 63].

The associated rotated expansion coefficients $\mathbf{P}_{R,s}$ are then obtained by rotating the original expansion coefficients $\mathbf{P}_s$:

$$\mathbf{P}_{R,x} = \mathbf{P}_x \Sigma^{-1/2} \mathbf{R} \mathbf{W}_s \quad (2.46)$$

For an orthogonal rotation, the rotated expansion coefficients remain uncorrelated:

$$\mathbf{P}_{R,x}^H \mathbf{P}_{R,y} = \left(\mathbf{P}_x \Sigma^{-1/2} \mathbf{R} \mathbf{W}_x \right)^H \left(\mathbf{P}_y \Sigma^{-1/2} \mathbf{R} \mathbf{W}_y \right) \quad (2.47)$$

$$= \mathbf{W}_x \mathbf{R}^H \Sigma^{-1/2} \mathbf{P}_x^H \mathbf{P}_y \Sigma^{-1/2} \mathbf{R} \mathbf{W}_y \quad (2.48)$$

$$= (N - 1) \mathbf{W}_x \mathbf{W}_y. \quad (2.49)$$

while the unitary property of the rotated pattern vectors is removed:

$$\mathbf{V}_{R,x}^H \mathbf{V}_{R,y} = \left(\mathbf{V}_x \Sigma^{1/2} \mathbf{R} \mathbf{W}_x \right)^H \left(\mathbf{V}_y \Sigma^{1/2} \mathbf{R} \mathbf{W}_y^{-1} \right) \quad (2.50)$$

$$= \mathbf{W}_x \mathbf{R}^H \Sigma^{1/2} \mathbf{V}_x^H \mathbf{V}_y \Sigma^{1/2} \mathbf{R} \mathbf{W}_y \quad (2.51)$$

$$= \mathbf{W}_x \mathbf{R}^H \Sigma \mathbf{R} \mathbf{W}_y. \quad (2.52)$$

A drawback of the proposed rotation procedure—and which has apparently gone unreported as far as we know—is that the total squared covariance is *not* conserved by the rotation which makes Equation (2.37) invalid for estimating the relative importance of the rotated modes. We provide a proof of this in the Appendix A.1.

Beyond rotation for better interpretability, Hilbert MCA can be applied to capture the temporal evolution of coupled patterns, akin to Hilbert PCA (see Section 2.1.3). This is especially useful for analyzing propagating patterns like the MJO, which involve covarying patterns in different climate variables, such as wind speed in the troposphere and lower stratosphere, and precipitation. By augmenting data with the Hilbert transform, we obtain complexified fields and solve the maximization problem (Equation (2.31)) for these augmented data, capturing the temporal evolution of coupled patterns. It seems that Hilbert MCA has seen limited application in climate science [e.g. 41]. In this thesis we utilize the combined power of Hilbert and rotated MCA to identify and interpret coupled patterns of variability in the climate system.

PCA, MCA and CCA MCA reduces to PCA when both sets are identical. Furthermore, MCA can be viewed as two PCAs followed by rotation to align the results of the two analyses as

closely as possible [28]. MCA is also closely related to Canonical Correlation Analysis (CCA), which, instead of maximizing shared covariance, seeks patterns that maximize correlation. Unlike MCA, where pattern vectors are by construction orthogonal, yielding correlated expansion coefficients, CCA results in pattern vectors that are not necessarily orthogonal but produce uncorrelated canonical variates [85, 180]. In fact, PCA, MCA and CCA can all be seen as special cases of a more general framework [13, 160].

drawbacks of MCA according to Swenson: 1. Drawbacks of MCA are its representation of relationships with transformations that are necessarily orthogonal (Newman and Sardeshmukh 1995) 2. its lack of a least squares estimate because of nonzero correlation between modes (Tippett et al. 2008) 3. its bias toward modes with larger variance given that it maximizes squared covariance.

2.3 Neural Networks

2.3.1 Introduction to Neural Networks

Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. Basic Architecture: Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process.

2.3.2 Recurrent Neural Networks

Introduction to RNNs¹²: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. Challenges with RNNs: Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM¹³: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. LSTM Architecture: Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. Mathematical Formulation: Provide the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

¹²Graves 2021 Supervised Sequence Labelling with Recurrent Neural Networks

¹³Hochreiter & Schmidhuber 1997

2.3.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

Layerwise Relevance Propagation (LRP)

Concept of LRP¹⁴: Describe what LRP is and its importance in understanding neural network decisions. Mechanism of LRP: Explain how LRP works, including the propagation rules and the relevance score computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret model predictions.

simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables
simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables

SHapley Additive exPlanations (SHAP)

Introduction to SHAP¹⁵: Discuss the SHAP method, its foundation in cooperative game theory, and its advantages. SHAP Values: Explain how SHAP values are computed and their interpretation. Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to provide consistent and locally accurate explanations.

2.4 Gaussian Processes

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

2.4.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

¹⁴Bach et al. 2015

¹⁵Lundberg 2017

2.4.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. **Kernel Functions:** Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. **Hyperparameter Tuning:** Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

2.4.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. **GPs in Spatial Modeling:** Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. **Log Gaussian Cox Model:** Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. **Mathematical Formulation:** Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. **Integration with Log Gaussian Cox Model:** Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

2.4.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. **Inference and Prediction:** Explain how to perform inference and make predictions using the fitted model.

2.5 Comparative Perspective

Dimensionality Reduction Techniques in Climate Science: Bridging Theory and Practice

Over the past few decades, Earth system sciences have experienced an exponential increase in the volume and complexity of data [126], driven by the proliferation of Earth observations satellites [56], high-resolution climate models [66] and citizen science contributions [49]. As a result, researchers must now navigate the immense challenge of analyzing vast datasets to extract meaningful insights about Earth's climate system [150]. Dimensionality reduction techniques have become indispensable for addressing this challenge by simplifying complex data and revealing key underlying patterns.

In the previous chapter, we explored the theoretical foundations of several dimensionality reduction techniques commonly applied in Earth system sciences. We demonstrated how common PCA extensions can enhance MCA, yielding more physically interpretable patterns that reflect the shared dynamics between two datasets. This chapter we shift focus to the practical challenges that arise when applying these techniques to the increasingly large Earth system datasets and the solutions provided by open-source software. In particular, Section 3.3 will introduce the XEOFS Python library, which is designed to facilitate scalable dimensionality reduction with a particular focus on climate science. This package provides user-friendly access to both traditional and advanced methods, enabling researchers to conduct sophisticated analyses without needing extensive expertise in data science. This section is based upon the work presented in:

- **Rieger, N.** & Levang, S. J. xeoofs: Comprehensive EOF Analysis in Python with Xarray. *Journal of Open Source Software* **9**, 6060. doi:[10.21105/joss.06060](https://doi.org/10.21105/joss.06060) (2024)

3.1 Open-Source Tools for Scalable Climate Data Analysis

Climate science is inherently data-intensive, relying on the analysis of large, complex datasets to understand the Earth's climate system. However, traditional analysis methods often struggle

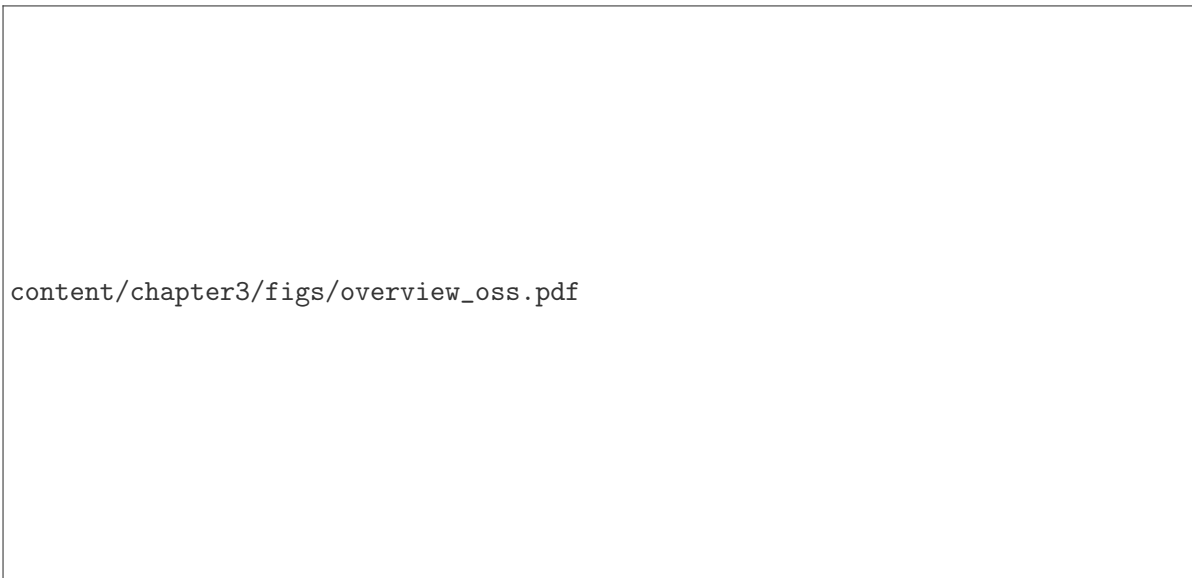


Figure 3.1 — Open-source software (OSS) projects in Earth system sciences. Panel A illustrates the number of new OSS projects created annually, highlighting the growing importance of Python-based projects. Panel B shows the relative share of programming languages used in all OSS projects since 2010, excluding languages with a share of $\leq 0.5\%$. The OSS project data is sourced from *Open Sustainable Technology* (OST) [5]. For this analysis, Earth system science projects include the OST categories *Atmosphere*, *Climate Change*, *Cryosphere*, *Hydrosphere* and *Biosphere*. Data for 2024 includes only the first half of the year, up to August.

to handle the scale and complexity of modern climate data, making the application of these techniques increasingly difficult [45]. This creates a growing need for tools that lower the barrier to entry for non-experts, allowing them to manage and analyze vast datasets effectively.

One of the central challenges in climate science today is the "data deluge" – the overwhelming volume of climate data and the difficulty of extracting useful insights from it. As Schnase *et al.* [151] noted, 'data analytics, the capacity to perform useful scientific analyses over large quantities of data in reasonable amounts of time' is a critical challenge for climate science. Without effective tools to transform large datasets into actionable knowledge, the mission of climate science falters.

To ensure that data analysis remains accessible to the broader scientific community, free, robust, and scalable software tools are essential [39]. These tools must allow researchers to work with large datasets, even without specialized programming skills, and simplify the processing of data produced by climate models.

It therefore comes as no surprise that the open-source software (OSS) movement has gained significant traction in the scientific community (Figure 3.1), with researchers increasingly recognizing its benefits for fostering collaboration, enhancing transparency, and ensuring reproducibility in research [83, 92, 118]. Python, in particular, has become a preferred language

for scientific computing (Figure 3.1 B), presumably due to its ease of use, extensive library support and active developer community. Notable examples of successful Python-based OSS projects include NUMPY [64, 124], SCIPY [124, 171] and MATPLOTLIB [79] libraries, which have become essential tools for scientific computing and visualization contributing to the democratization of scientific tools, driving innovation and facilitating new scientific discoveries [156].

The *Pangeo* project [127] has been especially influential in promoting open-source tools for big data geoscience. Two key components of the Pangeo ecosystem are the XARRAY [75] and DASK [34] Python libraries. XARRAY enhances the user experience by introducing labeled data structures that simplify the manipulation of multidimensional arrays, while DASK enables parallel and distributed computing. Together, these tools have revolutionized the way climate researchers handle large datasets, allowing them to focus on scientific questions rather than technical hurdles.

3.2 Challenges in Large-Scale and Multidimensional Climate Data Analysis

Analyzing climate data often involves multivariate statistical techniques to identify patterns within large, multidimensional datasets. In Section 2, we only explored a small subset of available methods for pattern identification in climate science [see e.g. 63, 77]. Most existing techniques assume that data can be structured as a two-dimensional (2D) matrix. However, climate data, whether from Earth observations or simulations, often exhibit two key characteristics:

1. **Multi-dimensionality:** Climate data span several dimensions, including spatial (longitude, latitude, height), temporal (time steps, forecast lead times), and categorical (variables, sensors).
2. **Large Volume:** These datasets are often so large that they exceed the memory capacity of a single computer and even of high performance computing (HPC) infrastructures.

These characteristics have led to the widespread adoption of tools like XARRAY and DASK which facilitate the handling of large, multidimensional datasets in the Python ecosystem.

However, applying dimensionality reduction techniques such as PCA and MCA (Section 2) presents several challenges. Climate datasets typically require significant pre-processing, including variable weighting (Section 2.1), removing invalid grid points (e.g., land cells in sea surface temperature data), and reshaping multidimensional arrays into 2D matrices. After analysis, researchers must reverse this process – transforming the 2D results back into

their original multidimensional form, reinserting missing data (such as land grid cells), and removing weights. This workflow can be cumbersome, error-prone, and time-consuming.

Earlier tools, such as the EOFs and XMCA libraries, addressed PCA and MCA applications in climate science. However, they lack full compatibility with XARRAY and DASK limiting their scalability and effectiveness when dealing with large datasets. For example, these libraries do not support XARRAY's Dataset object, and neither leverages randomized linear algebra techniques [43, 60], which can significantly improve computational performance for large datasets.

In addition, researchers often need more than dimensionality reduction. Assessing the statistical significance of the extracted patterns, for example, requires bootstrapping techniques [40], adding further complexity to the analysis process. Implementing these advanced methods often requires substantial coding expertise, which can be a barrier for many scientists [181].

Moreover, several classical methods developed over the years for pattern extraction in climate science have not yet been implemented in Python or integrated with XARRAY and DASK. This lack of availability limits researchers' ability to apply these well-established techniques to their datasets.

3.3 The XEOFS library: A Solution For Dimensionality Reduction in Climate Science

The XEOFS library [144] was developed to address these challenges by providing a Python package for applying dimensionality reduction techniques in climate science. Built with compatibility in mind, XEOFS integrates seamlessly with XARRAY and DASK, making it suitable for large, multidimensional datasets. The library offers a user-friendly interface to both traditional and advanced methods commonly used in climate data analysis (Table 3.1). Its development adheres to key principles of open-source, scientific programming [6, 136], including clean coding practices, rigorous test design [80], version control [131], robustness [164], usability [107], and thorough documentation¹ [97]. The library also follows best practices in open-source software development. For Python specifically, XEOFS aligns with the guidelines set forth by the Pangeo project [127] and the *pyOpenSci* Packaging Guide [174].

Key features of the XEOFS library include:

Multi-dimensional & Labeled Data Designed specifically for XARRAY objects, XEOFS retains data labels and dimensional structures throughout the analysis, which helps to maintain clarity and to reduce errors.

¹The XEOFS online documentation is available at <https://xeofs.readthedocs.io>

Table 3.1 — Overview of available methods in the XEOFS library [144] (as of September 2024) for single-set (one dataset), cross-set (two datasets), and multi-set (more than two datasets) analysis. PCA: Principal Component Analysis; POP: Principal Oscillation Pattern; OPA: Optimal Persistence Analysis; CCA: Canonical Correlation Analysis; RDA: Redundancy Analysis; MCA: Maximum Covariance Analysis; CPCCA: Continuum Power CCA. Rotation techniques include orthogonal varimax [87] and oblique promax [69]. Combinations like rotated Hilbert PCA [71] are also supported. Multi-set CCA is based on the MCCA class of the CCA-ZOO library [23] and Sparse PCA uses the RISTRETTO library [43].

	METHOD	REFERENCE
SINGLE-SET ANALYSIS	PCA	[74, 108, 128]
	Complex PCA	[93, 158]
	Hilbert PCA	[7, 72, 137]
	Rotated PCA	[72, 140]
	Extended PCA	[15, 175]
	GWPCA	[65]
	Sparse PCA	[44]
	POP	[68]
	OPA	[35, 36]
CROSS-SET ANALYSIS	CCA	[14, 74]
	RDA	[158]
	MCA	[14, 158]
	CPCCA	[160]
	Complex CCA	
	Complex RDA	
	Complex MCA	
	Hilbert CCA	
	Hilbert RDA	
	Hilbert MCA	[41]
	Rotated MCA	[24]
	Rotated CPCCA	
MULTI-SET ANALYSIS	CCA	[88]
	Regularized CCA	[170]

Out-of-Memory Computations Leveraging DASK, XEOFS can process large datasets that exceed the memory capacity of a single machine or HPC systems.

Computational Efficiency By employing randomized linear algebra, XEOFS dramatically improves the efficiency of dimensionality reduction techniques, particularly for large datasets. Benchmark comparisons show that XEOFS is approximately an order of magnitude faster than EOFs for medium to large datasets (Figure 3.2).

Additionally, XEOFS offers several auxiliary features:

Handling Missing Values The library can manage typical cases of missing data, such as masked ocean or land cells.

Weighted Analysis XEOFS supports weighted analyses, allowing researchers to quickly apply common weighting schemes (e.g. standard deviation or latitude-based cosine weighting, see Section 2.1). It also supports custom weighting schemes.

Significance Testing A built-in bootstrapping interface enables users to test the significance of extracted modes, reducing the need for additional coding.

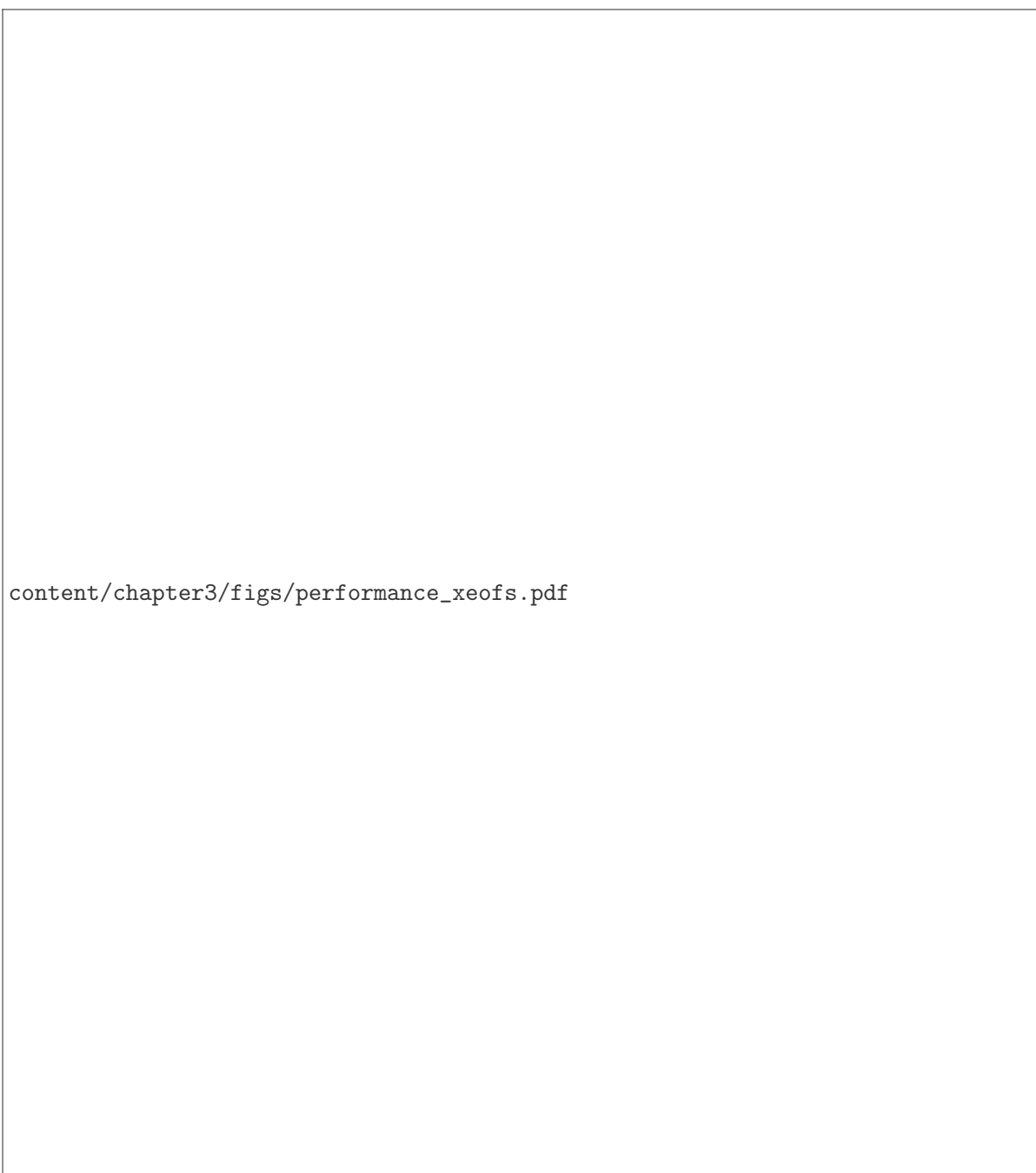
Model Serialization Trained models can be saved and loaded in HDF5, netCDF4 or cloud-optimized Zarr [117] formats ensuring compatibility with standard climate data formats [67].

The XEOFS library is modular and object-oriented, allowing for future extensions and integration of new dimensionality reduction methods. It provides an intuitive application programming interface (API) similar to the popular machine learning SCIKIT-LEARN library [129] that simplifies the application of these techniques, even for researchers with limited programming experience.

The library is available on the Python Package Index (PyPI) and can be installed via pip and conda. Its source code is hosted on GitHub, where users can contribute to its development and report any issues. Licensed under the MIT license, XEOFS is free to use and modify for both academic and commercial purposes.

3.4 Conclusion

The development of the XEOFS library represents a contribution toward addressing the challenges of scalable and efficient dimensionality reduction in climate data analysis. However, there are several opportunities for further refinement and expansion to better serve the needs of the climate science research community.



content/chapter3/figs/performance_xeofs.pdf

Figure 3.2 — Computational performance of PCA using the XEOFS library (version 3.0.0). Panel A displays computation times (in seconds) for varying dataset sizes, with different numbers of observations (samples) and variables (features). Panel B illustrates the speedup achieved by XEOFS compared to the EOFS library (version 1.4.1). Panel C presents the performance of delayed (lazy) Dask computation versus eager computation in both small data ($n = 10$ samples) and large data ($n = 10,000$ samples) scenarios. The results demonstrate a significant efficiency gain for medium to large datasets (greater than 1 MB) where XEOFS outperforms EOFS by approximately an order of magnitude. Additionally, XEOFS's Dask implementation for chunked out-of-memory data offers further improvements for larger datasets. Interestingly, this efficiency gain is observed primarily for datasets with a balanced number of features and samples, while performance does not significantly improve for extremely "tall-and-skinny" or "short-and-wide" matrices. Notably, the Dask implementation in EOFS performs worst on larger datasets. For small datasets (less than 1 MB), the Python overhead is more pronounced, with EOFS being slightly more efficient, although the difference is generally less than a factor of two. The benchmark was conducted on a system with Dual Intel Xeon Gold 6248R 24C 3.0 GHz processors and 192 GB of RAM, running Rocky Linux 8.

Future updates to XEOFS could introduce additional state-of-the-art methods that have recently gained attention. Nonlinear dimensionality reduction techniques, such as those proposed by Bueso *et al.* [17] as well as spectral methods as proposed by Guilloteau *et al.* [59], would enable researchers to capture more complex relationships within climate datasets. Incorporating approaches like nonlinear PCA, kernel-based methods [152], and advanced sparse algorithms to improve interpretability [58, 154] would provide users with a broader toolkit to explore the intricate dynamics often present in climate data.

Another area for improvement is the expansion of bootstrapping techniques for evaluating the statistical significance of extracted patterns, particularly in MCA and rotated PCA. Bootstrapping helps researchers assess the robustness of their findings, distinguishing meaningful patterns from noise. By integrating more advanced bootstrapping algorithms, such as those discussed by Timmerman *et al.* [167], XEOFS could simplify the workflow for significance testing, making these analyses more accessible without requiring additional coding expertise.

The next chapter will focus on applying dimensionality reduction techniques to real-world climate datasets, highlighting the practical use of XEOFS. These applications will demonstrate how the package can help extract meaningful insights from complex datasets. by exploring interactions between the ocean and atmosphere.

Part II

Case Studies

Shared Variability between Temporally Lagged SST and Continental Precipitation 4

The Earth's climate system is highly complex, with various components interacting across different spatial and temporal scales. A key aspect of these interactions is atmospheric teleconnections—large-scale patterns of pressure and circulation anomalies that link climate and weather conditions across distant regions, often thousands of kilometers apart. These teleconnections enable climate variations in one area to influence weather and climate in far-reaching locations. For example, the El Niño-Southern Oscillation (ENSO) significantly affects global weather patterns, altering rainfall and temperature distributions far from the Pacific Ocean where it originates [145, 149]. Similarly, the North Atlantic Oscillation (NAO) impacts weather in Europe and North America.

A critical feature of atmospheric teleconnections is their lagged nature; the influence of a climatic anomaly in one region on another does not occur instantaneously but with a temporal delay. For instance, the atmospheric response to ENSO typically manifests several months after the peak of an El Niño event, with the maximum response in the tropical North Atlantic occurring approximately five months later [42, 89, 148]. Understanding these delayed relationships is essential for enhancing the predictability of climate variability, thereby improving climate forecasts from sub-seasonal to decadal timescales [157, 184]. Additionally, incorporating lagged teleconnections into climate models leads to more accurate simulations and projections by accounting for previously unrecognized interactions.

While teleconnections have been extensively studied across various timescales—from intraseasonal [e.g. 157, 184] to decadal [e.g. 104]—many studies have employed hypothesis-driven approaches. There is a growing need for data-driven, exploratory analyses that can identify dominant patterns of shared variability without prior assumptions. This chapter addresses this need by examining the shared variability between sea surface temperatures (SST) and continental precipitation across seasonal to decadal timescales. We utilize rotated Hilbert Maximum Covariance Analysis (MCA; see Section 2.2) to uncover and interpret the dominant patterns of

covariability, implicitly capturing lagged teleconnections through the Hilbert transform and enhancing interpretability through mode rotation.

Preliminary results from this research were published in

- **Rieger, N. et al.** Lagged Teleconnections of Climate Variables Identified via Complex Rotated Maximum Covariance Analysis. *Journal of Climate* **34**, 9861–9878. doi:[10.1175/JCLI-D-21-0244.1](https://doi.org/10.1175/JCLI-D-21-0244.1) (2021),

demonstrating the effectiveness of Hilbert MCA in identifying lagged teleconnections between SST and precipitation. This thesis chapter extends those findings in several significant ways. First, it expands the spatial and temporal scope of the analysis by incorporating newly available data. Second, it introduces a more objective method for determining the number of rotated modes, thereby increasing the robustness of the results. Finally, it explores the implications of using correlation instead of covariance as a metric for shared variability, utilizing and extending the Continuous Power Canonical Correlation Analysis (CPCCA) framework introduced by Swenson [160].

The structure of this chapter is as follows: We begin by describing the data sources and methodological framework (Section 4.1), with a particular focus on the data preprocessing and analysis decisions. Next, we present the results, highlighting the first ten rotated modes of shared variability between SST and continental precipitation (Section 4.2). These findings are then discussed in relation to existing climate indices and known teleconnection patterns, highlighting both consistencies and novel insights. Finally, we conclude with a summary of the key findings and their implications for future climate research and forecasting (Section 4.3).

4.1 Methods & Data

We use the fifth-generation ERA5 reanalysis dataset provided by the European Centre for Medium-Range Weather Forecasts (ECMWF) [32] to investigate the shared variability between sea surface temperature (SST) and continental precipitation on seasonal to decadal time scales. ERA5 offers global coverage on a regular longitude-latitude grid with a spatial resolution of 0.25° and monthly temporal resolution. Spanning from 1940 to the present, it allows us to capture long-term climate variability, trends, and potential teleconnections between SST and precipitation. ERA5 has proven to be a reliable source for climate studies due to its high spatial resolution and strong agreement with observations [70, 155, 183]. Precipitation is particularly well represented in the extratropics of the Northern Hemisphere, where most continental landmass resides, making it a suitable proxy for observational data. However, the representation of precipitation in the tropics is less accurate [96], which we need to consider when discussing dominant patterns of variability in tropical precipitation. Nevertheless, the

recent extension of ERA5 back to 1940—which we use here—has significantly improved the representation of precipitation during the 1950s and 1960s [155].

Our geographical focus is between 60°S and 70°N because the majority of landmasses experiencing mostly liquid precipitation lie within these latitudes. Excluding Antarctica helps us avoid biases due to its extreme dryness and sparse observational data. Additionally, SST variability is less reliable in polar regions because of partial sea ice coverage throughout the year. The Southern Hemisphere, in general, has historically fewer observations, affecting data reliability [155].

We concentrate on continental precipitation by masking out oceanic precipitation using the land-ocean mask at a 1:110 million scale provided by Natural Earth¹. To emphasize teleconnections operating on annual to decadal time scales, we apply a centered 7-month moving average to both the SST and precipitation time series. This approach removes high-frequency variability, such as monsoonal circulations, while retaining low-frequency phenomena like ENSO and PDO. Importantly, we retain the seasonal cycle to test whether we can recover dominant modes of variability using rotated Hilbert Maximum Component Analysis (MCA) (see Section 2.2) despite its presence in the data.

Our study aims to identify teleconnections—connections between SST and continental precipitation in regions that are not necessarily geographically close. However, local connections between SST and precipitation are common; for example, coastal precipitation and river plumes are related to changes in SST [e.g. 101]. Since ERA5’s high spatial resolution (approximately 30 km at the equator) can resolve these processes, it’s likely that an MCA of both datasets might also detect these local-scale phenomena. To mitigate the influence of small-scale processes and focus on larger-scale teleconnections, we coarsened the data by applying spatial averaging from a 0.25° to a 1.00° longitude-latitude grid, effectively smoothing out localized patterns. This processing step left us with 36,588 and 16,920 non-masked grid cells (features) for SST and continental precipitation, respectively. Additionally, to counter the effect of grid size distortion toward the poles [123], we applied latitude weighting to the data as described in Section 2.1.

To detect lagged covariance between SST and precipitation, we applied the Hilbert transform to each time series individually, converting them into complex-valued datasets. To minimize boundary effects and spectral leakage caused by the Hilbert transform (see Section 2.1), we first extended each time series in both directions using an exponentially decaying function with a three-month e-folding time. This extension smooths the data, preventing abrupt jumps and aligning the decay with the dominant seasonal cycle in our dataset. As a result, each time series became three times longer than the original. After applying the Hilbert transform to the extended series, we trimmed the extra sections, retaining only the original time series within the complex data. This approach helps prevent the Hilbert transform from artificially inflating

¹naturalearthdata.com

the data's variance, which could lead to spurious correlations in the Hilbert MCA analysis [143].

Handling the large cross-covariance matrix—in the order of $10^4 \times 10^4$ —poses significant computational challenges. To address this, we performed Hilbert MCA in the principal component (PC) space [barnett_origin_1987]. We first transformed each dataset into its principal axes and retained enough principal components to explain 99 % of the variance in each dataset. This reduction resulted in 74 components for SST and 199 components for continental precipitation, shrinking the cross-covariance matrix to a more manageable size of 74×199 . By decreasing the dimensionality, we not only made the computations feasible but also stabilized the solution by regularizing the initially ill-conditioned problem.

With the dimensionality reduced, we proceeded to perform MCA on the complexified, PC-transformed data to identify shared variability patterns between SST and continental precipitation. However, standard MCA introduces two main challenges. First, the spatial patterns generated are dense, meaning that nearly all grid points contribute to each pattern, which complicates interpretation. Second, the expansion coefficients (time series) are uncorrelated with one another. While this lack of correlation simplifies the mathematics, it may not accurately reflect the reality of geophysical processes, which are typically interconnected and influence each other.

To address these issues, we applied an additional rotation to the MCA solution, following the methodology of Cheng & Dunkerton [24] (Section 2.1). Before performing this rotation, we needed to make two key decisions: (1) determining the optimal number of modes to rotate, and (2) choosing between orthogonal or oblique rotation.

How many modes to rotate? Selecting the right number of modes to rotate involves balancing the inclusion of significant shared variability while minimizing the incorporation of noise from higher-order modes. Rotating too few modes may overlook important patterns, whereas rotating too many can introduce irrelevant noise. Additionally, we sought a solution where adding more modes would not significantly alter the results, ensuring stability and reliability.

In MCA, each mode's significance is assessed using the squared covariance fraction (SCF), which measures the proportion of shared variability each mode explains (Equation ??). Within our PC space, the total shared variability is represented by 74 modes. Analysis of the SCF for the initial non-rotated modes (Figure 4.1 A) revealed that nearly all squared covariance is captured within the first six modes, with the SCF dropping below 0.01 % for subsequent modes. This rapid decline can also be understood by the fact that *squared* covariance is measured.

To ensure a stable rotation solution, we employed an objective criterion to determine the optimal number of modes to rotate. We iteratively rotated the first $k \in [10, 40]$ modes, resulting in 31 unique rotation solutions. For each solution, we computed the amplitude patterns of

content/chapter4/figs/raster/tele_singular_spectrum.png

Figure 4.1 — **Caption** B. the sum of squared singular values of the whitened cross-covariance matrix is a conserved quantity in (Hilbert) CPCCA, but not under rotation [24]. C. The sum of singular values (whitened covariance) is not in general a quantity conserved within the CPCCA framework. It is conserved, however, under orthogonal rotation. Here, since we use an oblique promax rotation, the cumulative singular values are not conserved. D. mode stability index χ shows that the highest stability is obtained when 23 modes are rotated which higher modes do not change the solution significantly. E. correlation matrix of expansions coefficients indicates that a weak but noticable correlation between mode 1 and 2 is present. The squared covariance fraction is computed using the residual formulation (see equation???)

the complex-valued singular vectors for the first 10 modes of both SST and precipitation. A stable solution is characterized by amplitude patterns that remain consistent across different rotations, even if the order of modes varies slightly due to the diminishing SCF in higher modes (Figure 4.1 A). We assumed that the most stable and physically meaningful modes would be among the first 10.

To evaluate the stability of these rotation solutions, we measured the similarity between amplitude patterns using cosine similarity—a non-centered version of the Pearson correlation coefficient, also known as Tucker’s coefficient of congruence. The cosine similarity ranges from -1 to 1 , where 1 indicates identical patterns, 0 indicates orthogonal patterns, and -1 indicates opposite patterns. For each of the 31 rotation solutions, we compared the first 10 amplitude patterns against those of the other solutions. Patterns were considered identical if their cosine similarity exceeded 0.9 . We then calculated the mode stability index, χ , as the fraction of modes that remained consistent across all solutions. Figure 4.1 D shows that the highest stability is achieved when rotating 22 modes, as adding more modes did not significantly change the results. Therefore, we focused our analysis on the solution obtained by rotating the first 22 modes.

$$s_{k,k',i} = \max_j \left(\frac{\mathbf{p}_{k,i}^\top \mathbf{p}_{k',i}}{\|\mathbf{p}_{k,i}\| \|\mathbf{p}_{k',i}\|} \right) \quad (4.1)$$

Choosing Between Orthogonal and Oblique Rotation The choice between orthogonal and oblique rotation involves a trade-off between interpretability and complexity. Oblique rotations can produce simpler spatial structures—that is, patterns restricted to fewer grid points—resulting in more localized and interpretable modes [141]. Orthogonal rotations keep the expansion coefficients uncorrelated, which simplifies their interpretation but may not reflect the interconnected nature of geophysical processes. Oblique rotations allow for correlated modes, potentially capturing more realistic relationships but adding complexity to the interpretation.

Given this, we initially apply an oblique rotation to the MCA solution and consider using orthogonal varimax rotation if the obtained modes exhibit only weak correlations (absolute pairwise correlation coefficients below 0.2). For the initial rotation to assess the degree of correlation between the expansion coefficients, we rotate the first 22 modes using the promax rotation scheme [69] with a weak rotation power of $p = 2$, as recommended by Richman [141]. Computing the absolute pairwise correlation coefficients between the first 10 promax-rotated, complex-valued expansion coefficients yields Pearson correlation coefficients below 0.14 , suggesting only weak correlations between the leading modes. Consequently, we choose to proceed with the orthogonal rotation case (promax rotation power $p = 1$). The results of the oblique rotation are presented in the appendix (Figure B.4–B.13).

Indices of known Climate Oscillations For comparative purposes, we obtained climate indices of monthly means from the National Oceanic and Atmospheric Administration (NOAA; <https://psl.noaa.gov/data/climateindices/list/>), including the Oceanic Niño Index (ONI) provided by the NOAA Climate Prediction Center, the PDO as defined by Mantua & Hare [111], and the Atlantic Meridional Mode (AMM) as defined by Chiang & Vimont [30]. We calculated the ENSO Modoki Index (EMI) following Ashok *et al.* [3] using the formula $EMI = R_A - \frac{1}{2}R_B - \frac{1}{2}R_C$, where R denotes the area-averaged ERA5 SST anomalies over regions A (165°E-140°W; 10°S-10°N), B (110°W-70°W; 15°S-5°N), and C (125°E-145°E; 10°S-20°N). Additionally, we define an Oceanic Warming Index (OWI) as the area average of ERA5 SST anomalies between 60°S and 70°N, using the same latitude weighting as in our MCA data preparation. All indices are smoothed using a centered 7-month moving average to align with our data preprocessing and are restricted to the period from 1940 to 2022. We detrend all indices except the OWI using linear regression and further center and standardize them for better comparison with our obtained expansion coefficients.

Power Spectral Density We compute the power spectral density (PSD) for the first 10 expansion coefficients of SST and continental precipitation, as well as for the climate indices. For the complex-valued expansion coefficients, we extract the real part after phase-shifting the coefficients so that, for each mode, the phase at the location with the maximum absolute value of the SST heterogeneous patterns is zero. This step is necessary because the complex-valued singular vectors from our Hilbert MCA are defined only up to a phase shift. By aligning the phase this way, the real part represents the maximum fraction of the lagged variability and captures the temporal evolution of the strongest signal in the data. Prior to PSD computation, all time series are linearly detrended. The PSD is computed using Welch's Overlapped Segment Averaging (WOSA) method [e.g., 130, Section 8.8], with 256 data points per segment and a Hanning window with 50 % overlap, resulting in a total of six segments. All PSDs are normalized to represent the fraction of energy density per frequency band.

Approximate confidence intervals for PSDs are often computed using the χ^2 distribution; however, this approach is valid only for a particular frequency and isn't directly applicable across multiple frequencies or the entire frequency range [cf. 135, p. 468]. Since we don't know in advance which frequencies are of interest, we employ the circular moving block bootstrap [133] to estimate confidence intervals for the PSDs. We generate 10 000 bootstrap samples by resampling the original time series with replacement, using a block length of 9 years. This choice balances the need for sufficient independent bootstrap samples with the requirement for long enough blocks to resolve the frequencies of interest. Note that this choice limits our ability to resolve periodicities longer than 9 years because we effectively scramble the autocorrelation function beyond this time lag. To preserve the annual cycle, we only consider January as a valid starting point for blocks. We compute the PSD for each bootstrap sample, and the 2.5th

and 97.5th percentiles of the bootstrap PSDs are used as the lower and upper bounds of the confidence intervals.

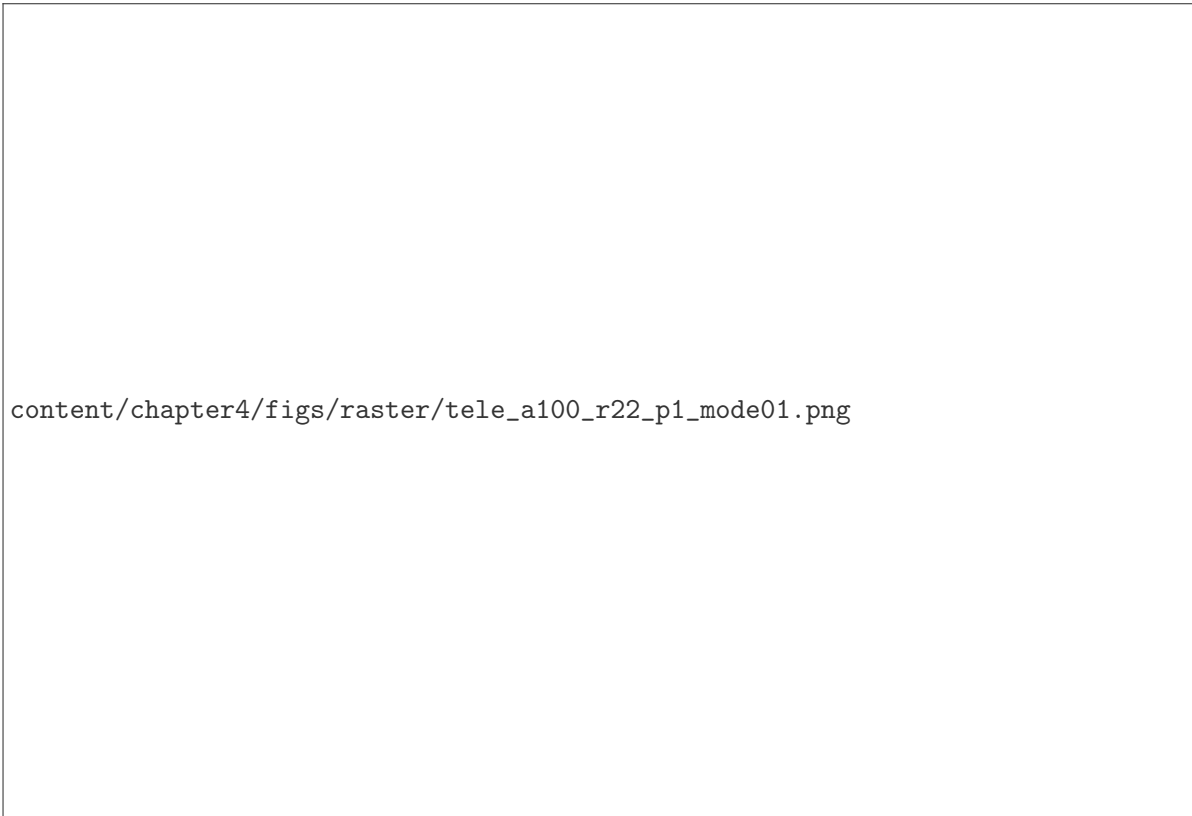
Since climate data often exhibit a ‘red noise’ spectrum, we compare the PSDs of the expansion coefficients against a theoretical ‘red noise’ spectrum [e.g., 180, Section 10.3]. We estimate the ‘red noise’ spectrum using a first-order autoregressive process (AR(1)) and compute the theoretical PSD using the formula given by Gilman *et al.* [53]. The autoregressive coefficient ρ is estimated from the data as the lag-1 autocorrelation coefficient. We then test the null hypothesis by comparing the spectral density of the data to that of the AR(1) process using the F-statistic at a significance level of $\alpha = 0.01$.

4.2 Results and Discussion

The rotated modes exhibit a more gradual decrease in cumulative squared covariance fraction (SCF) compared to the unrotated solution (Figure 4.1 A), with each of the first 12 modes explaining more than 0.1 % of the total squared covariance. A significant drop in SCF is noticeable after the 10th mode, where higher modes explain considerably less and decrease exponentially, indicating they may represent noise. Therefore, we focus our discussion on selected modes within the first 10. It is important to note that neither the cumulative SCF nor the sum of squared singular values are conserved under rotation (Figures 4.1 A, B). Only in the special case of orthogonal varimax rotation is the sum of singular values conserved (Figure 4.1 C).

Mode 1 – The Seasonal Cycle Mode 1 accounts for more than 99 % of the squared covariance between SST and continental precipitation (SCF1 = 99.1 %) and exhibits a clear one-year periodicity, unequivocally representing the annual cycle (Figures 4.2, 4.10 A). This mode shows a very strong link between SST and precipitation, evidenced by a Pearson correlation coefficient of $\text{rec}, 1 = 0.99$ between their expansion coefficients. As expected, the annual cycle manifests globally in both variables, with strong expressions in the extratropics for SST and within the tropics for precipitation. Exceptions to this pattern occur in transition zones, such as north of the Arabian Sea, where precipitation shifts between summer and winter-dominated regimes over a narrow region, or in areas with multiple rainy seasons like central East Africa.

The phase functions (Figures 4.2 C, D) correctly separate the seasonal cycle according to the timing of the annual peak, identifying, for example, the anticorrelation (phase shift of approximately π) between the northern and southern oceans. In SST, we observe that the eastern equatorial Pacific and the equatorial Indian Ocean exhibit a positive phase shift of $+\pi/4$ (about six weeks) relative to the dominant oceanic seasonal cycle. On the continents, the phase function illustrates the division between January-February (light blue) and July-August



content/chapter4/figs/raster/tele_a100_r22_p1_mode01.png

Figure 4.2 — Caption Amplitude and spatial phase is only plotted where p-values adjusted via the Benjamini-Hochberg correction for multiple testing are below 0.05. In addition, spatial phase function is masked out for regions where the spatial max-normalized amplitude is below 0.125 to remove ‘noisy’ phase values and for the sake of simplified visualization. Also to aid visualisation, the spatial phase function is binned into sections of 45° , or $\frac{1}{4}\pi$, the squared covariance fraction is computed using the residual formulation (see equation??), the explained variance is computed using the residual formulation, corr denotes the pearson correlation coefficient.



content/chapter4/figs/raster/tele_a100_r22_p1_mode02.png

Figure 4.3 — Caption Description

(dark red) dominated rainfall systems, identifying corresponding transition zones such as over the South American rainforest and central North America. It also highlights interesting dynamical regions that stand out from their surroundings, namely the west and east coasts of North America, the Mediterranean region, and, to a lesser extent, the East Asian monsoon region in China.

We also note that this first mode explains significantly more variance in SST (80.4 %) than in continental precipitation (62.1 %). This is expected because the seasonal cycle is a more dominant feature in SST, whereas precipitation is generally a noisier variable with a more complex spatiotemporal structure. Specifically, precipitation is more directly influenced by atmospheric variability, which exhibits high internal variability. Additionally, the ocean's thermal inertia allows it to absorb or release large amounts of heat with only minor temperature changes, making SSTs generally less variable than atmospheric variables. Finally, while SSTs change smoothly and slowly over vast spatial scales, continental precipitation is highly localized, so interannual variations in global circulation patterns can produce drastic changes in local precipitation patterns.

Mode 2 – ENSO Mode 2 ($SCF_2 = 3.9\%$, $rec_2 = 0.91$) is clearly associated with the ENSO phenomenon (Figure 4.3). The expansion coefficients of SST for this mode show a strong correlation with the ONI, with a Pearson correlation coefficient of $r_{2,ONI} = 0.92$, indicating a strong association with ENSO. Many of the resulting teleconnections illustrated in Figures 4.3 A–B have been previously reported and analyzed. For instance, during El Niño events, higher SSTs in the central and eastern Pacific and lower SSTs in the western Pacific positively correlate with increased rainfall along the west coast and southeastern coast of South America [165], the east and west coasts of North America [146], the East Asian monsoon region [178], and the Horn of Africa [84]. Simultaneously, precipitation decreases in northern South America [165], Oceania [33], South Africa [52], and the Indian monsoon region [26]. During La Niña events, these correlations are reversed. Recently, similar teleconnections have been identified using event coincidence analysis [179]. For a relatively recent summary of established ENSO-related rainfall patterns during El Niño and La Niña, see Lenssen *et al.* [102].

Furthermore, our results demonstrate the lagged dynamics between the core ENSO region in the western and central Pacific and other regions such as the Indian Ocean [91], the South China Sea [89], and the tropical North Atlantic [2, 29, 42, 148], consistent with previous studies. ENSO-related SST teleconnections in remote oceans often occur with delays: the Indian Ocean typically peaks 3 months, and the South China Sea and tropical North Atlantic about 5 months after the SST peak in the Pacific ENSO region [42, 89, 148]. The mechanism behind these lagged responses involves characteristic atmospheric circulation changes during El Niño events, leading to alterations in cloud cover and evaporation over remote oceans, resulting in increased net heat flux and SSTs [89, 95].

However, due to the broadband frequency spectrum of the SST expansion coefficients (Figure 4.10 B), with most energy between periodicities of roughly 2 and 5 years, the phase cannot be directly translated into a precise time shift. Assuming these periodicity limits, we can compute an approximate time shift. For example, the tropical Atlantic exhibits a phase shift of $+\pi/2$, translating into time shifts between 6 and 15 months, which is close to but slightly larger than the observed average time lag of 5 months. Nevertheless, the tropical North Atlantic clearly shows more positive phase values compared to the Pacific El Niño region, indicating it is positively time-shifted relative to the Pacific.

Recently, Scaife *et al.* [149] provided evidence for an even longer time lag—a one-year lagged extratropical response to ENSO—most pronounced in the North Atlantic. While they used sea surface pressure as the observational variable to assess the ENSO teleconnection, our SST-based analysis also indicates a weak signal of shared variability with ENSO in the North Atlantic, with a phase shift of approximately $+3\pi/4$. Using the periodicity limits from our analysis, this lag corresponds to time shifts between 9 and 22.5 months, aligning well with the reported one-year lagged response by Scaife *et al.* [149].

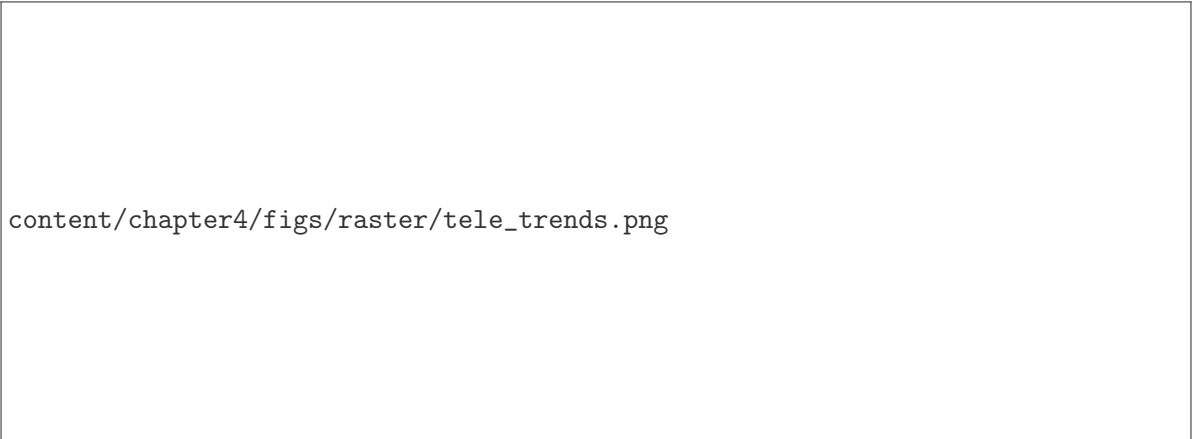
content/chapter4/figs/raster/tele_a100_r22_p1_mode03.png

Figure 4.4 — Caption Description

Considering all the lagged covariance collectively, ENSO accounts for approximately 5.8 % of SST variability and about 4.6 % of continental precipitation variability since 1940.

Mode 3 – Global Warming Mode 3 ($SCF_3 = 2.0\%$, $rec, 3 = 0.80$) represents changes in SST and the associated precipitation patterns due to global warming, as evidenced by the strong correlation with the OWI, $r_{OWI} = 0.78$ (Figure 4.4). The warming SST patterns are evident across all major ocean basins, with a clear spatial focus along the equator, except in the eastern equatorial Pacific (Figure 4.4 A). Additionally, amplitudes are slightly more pronounced in the Northern Hemisphere due to the asymmetric response of the northern and southern trade winds to global warming [182]. We also observe pronounced warming in the western parts of both the Pacific and Atlantic basins, regions associated with enhanced ocean heat transport [62, 182].

Similarly, we observe global trends in precipitation patterns, with the highest amplitudes in Central Africa, parts of Oceania, South and North America, and Central Asia (Figure 4.4 B). From the phase functions, we deduce an increase in precipitation (phase shift of zero) around tropical Oceania, while a decrease in precipitation (phase shift of $\pm\pi$) is suggested for Central Africa, central South Africa, western North America, and the Arabian Peninsula and sur-



content/chapter4/figs/raster/tele_trends.png

Figure 4.5 — Caption Description

rounding regions. These results are in strong spatial agreement with trends obtained from reanalysis data for both SST and continental precipitation (Figure 4.5), and largely concur with studies based on observational data [57] and, more recently, on CMIP5 climate simulations [54]. Collectively, these findings support the interpretation of this mode as representing the shared variability due to global warming.

However, some minor discrepancies exist between the obtained phase functions and the computed trends from reanalysis data. Notably, reanalysis data indicates strong positive precipitation trends of about 1 mm/decade in the eastern part of North America, which is not evident in our phase function. Additionally, the phase functions suggest an increase in precipitation in the southern parts of the Indian subcontinent, whereas reanalysis data shows a more complex pattern, with increases in the southeast and decreases in the western part.

Finally, it is important to emphasize that the expansion coefficients for both SST and precipitation only allow for meaningful interpretation of phases of zero (correlation) and $\pm\pi$ (anticorrelation), due to their broadband power frequency spectra (Figure 4.10 C). For phase shifts different from these values, no clear conclusions can be drawn—for example, in western Australia, where the phase shift is approximately $-\pi/4$.

Mode 4 – Nonlinear ENSO Modulation Mode 4 ($SCF_4 = 0.5\%$, $rec_4 = 0.82$) describes an oscillating SST anomaly mainly confined to the central Pacific (Figure 4.6), resembling the pattern popularized by Ashok *et al.* [3] as ‘El Niño Modoki’, represented by the EMI ($r_{EMI} = 0.56$). This pattern consists of higher SSTs in the central Pacific and lower SSTs in the eastern Pacific, which correlate with reduced precipitation in the East Asian monsoon region [47], Australia [163], parts of South America [165], and South Africa [139], and vice versa. Some of these teleconnections have recently been identified through event coincidence analysis [179], although our analysis suggests additional significant links to the East Asian monsoon



Figure 4.6 — Caption Description

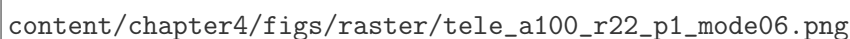


Figure 4.7 — Caption Description

region. Later research by Takahashi *et al.* [161], however, indicated that El Niño Modoki and the classical El Niño are essentially the same phenomenon, and that the perceived differences highlight the dynamic range of the system, specifically the nonlinearity of ENSO. Following this interpretation, the teleconnections identified in our analysis should not be considered a separate mode but rather a modulation of ENSO Mode 2 due to the system’s nonlinear behavior.

The power spectral density of this ENSO modulation shows a rather broadband energy distribution mostly indistinguishable from the theoretical ‘red noise’ spectrum, with the exception from some periodicities around 18 months (Figure 4.10 D). In particular, the energy distribution is markedly different from the periodicities typically associated with ENSO (2 to 7 years). Overall, the phase does not provide a clear interpretation apart from zero and $\pm\pi$ phase shifts.

Mode 6 – AMM Mode 6 ($SCF_6 = 0.2\%$, $rec_6 = 0.85$) is characterized by an SST pattern concentrated in the tropical and subarctic North Atlantic (Figure 4.7), explaining about 0.7 % of the overall SST variability. This pattern represents the AMM ($r_{AMM} = 0.75$), the dominant coupled ocean-atmosphere phenomenon in the tropical Atlantic [153]. The AMM’s impacts on precipitation—explaining about 0.8 % of total precipitation variability—in the African Sahel

content/chapter4/figs/raster/tele_a100_r22_p1_mode09.png

Figure 4.8 — Caption Description

zone, Central America, and northern South America are well documented [94, 113]. Only recently, Vittal *et al.* [172] found a link between the AMM and Indian summer monsoon rainfall, which they used to improve precipitation forecasts. Additionally, our results suggest further teleconnections between the AMM and precipitation variability over North America, the Mediterranean, and the East Asian monsoon region, offering potential for enhancing local rainfall predictions in these areas. Of particular interest is the southeastern United States, where a phase shift of $-3\pi/4$ —corresponding to a time lag of about 9 months assuming a periodicity of 2 years (Figure 4.10 E)—is observed.

Mode 9 – PDO Mode 9 ($SCF_9 = 0.2\%$, $rec_9 = 0.78$) exhibits a slow-oscillating SST pattern in the northeastern Pacific, correlating with localized precipitation patterns distributed across all continents, predominantly in North America (Figures 4.8 A, B). This spatial SST pattern is known as the PDO [112] ($r_{PDO} = 0.59$), a well-established climate index. Various processes originating in the tropics and extratropics have been proposed as the physical source of the PDO [122], with ENSO and PDO likely responding to the same forcing function [132]. Our analysis provides a means to disentangle ENSO- and PDO-related precipitation patterns, which are often similar in western North America [78]. Typically, the PDO exhibits strong energy around periodicities of 30 years; however, due to our preprocessing (using block sizes of 9

content/chapter4/figs/raster/tele_a100_r22_p1_mode10.png

Figure 4.9 — Caption Description

years), we cannot resolve such long periodicities in our analysis. Instead, we find higher-frequency, multi-year periodicities around 4 years (Figure 4.10 I), likely due to the significant feedbacks between ENSO and PDO modes [119, 162]. This finding is interesting because this spectral peak is not visible in the official PDO index, indicating slight differences in the composition of the PDO mode in our analysis compared to the traditional PDO index.

Mode 10 - IOD Mode 10 ($SCF_{10} = 0.2\%$, $r_{ec,10} = 0.83$) resembles the Indian Ocean Dipole (IOD) ($r_{IOD} = 0.49$), a coupled ocean-atmosphere phenomenon in the tropical Indian Ocean (Saji et al. 1999) that explains about 0.4% of the total SST variability. The SST pattern is characterized by a dipole structure with higher SST in the western and lower SST in the eastern Indian Ocean. The associated precipitation pattern is mainly concentrated in the surrounding tropical land masses, namely central east Africa and Oceania, explaining 1.2% of the total precipitation variability (Figure 4.9 A,B). The phase function shows a clear phase shift of $+\pi/2$, i.e. anti-correlation between precipitation patterns around the eastern and western Indian ocean, mirroring the dipole in the tropical Indian SSTs. Note that our results has only a weakly expressed dipole, with a much stronger center of attention in the eastern Indian ocean and much lower amplitude in the western Indian ocean (Figure 4.9 C, D). Although the PSD of this mode shows significant energy around 18-24 months compared to the ‘red noise’ model,

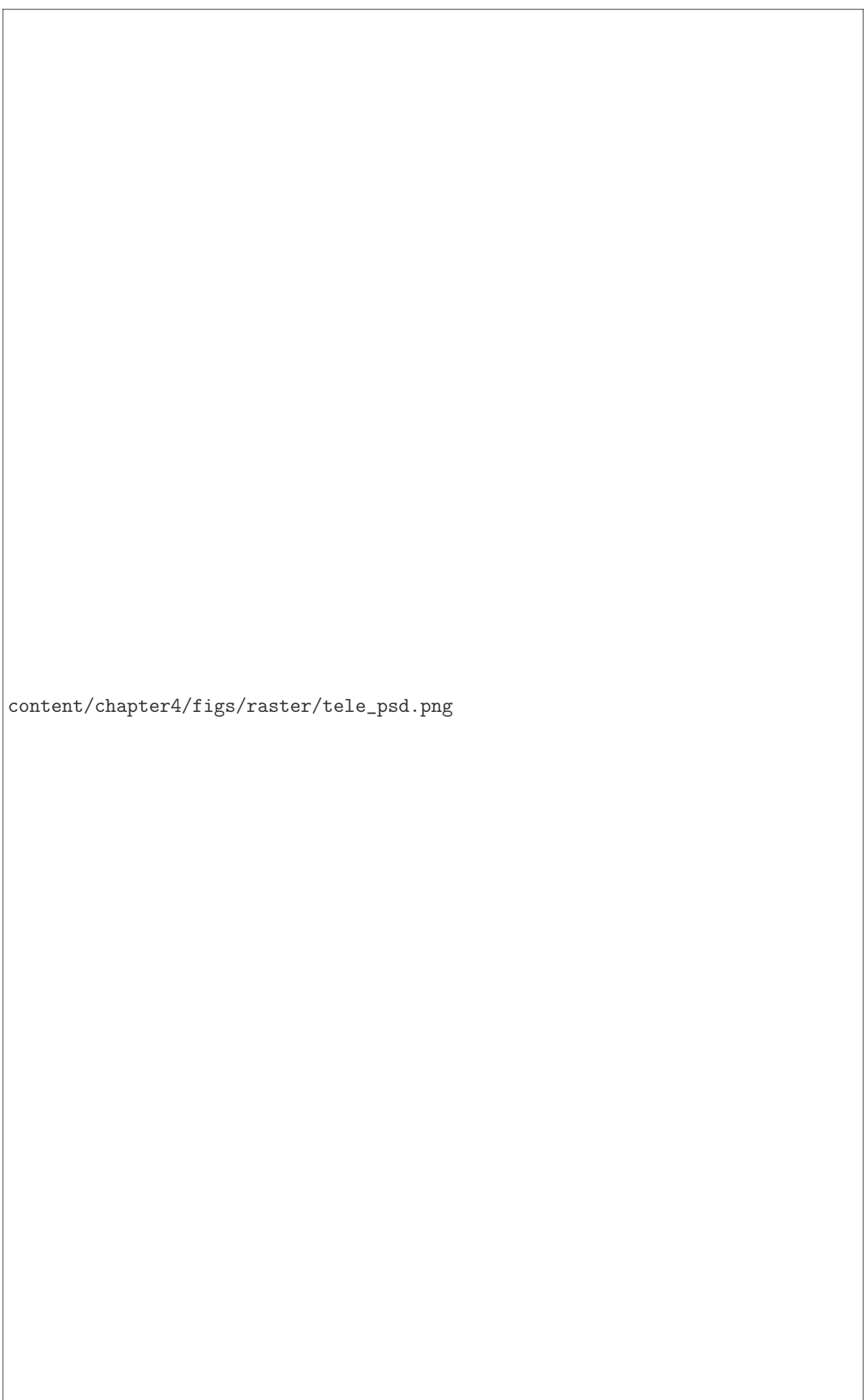
the spectral density remains relatively flat across a wide range of frequencies, reflecting the aperiodic and irregular nature of the IOD (Figure 4.10 J).

Other Modes of Shared Variability In our previous discussion, we focused on the dominant modes of shared variability identified by Rieger *et al.* [143]. With the expanded dataset used here, covering a broader spatial and temporal scope, we have now identified additional modes—specifically modes 5, 7, and 8—that bring new insights worth noting. Here, we provide a brief overview of these modes, pointing to some interesting preliminary observations while leaving detailed analysis for future work.

Mode 5 (Figure B.1) closely mirrors aspects of the atmospheric Pacific-South American (PSA) mode, which influences multiscale spatiotemporal patterns in the upper South Pacific Ocean. This mode significantly drives oceanic variability in the South Pacific subtropics, producing a potentially predictable climate signal connected to tropical regions [109]. This mode shows increased rainfall over southeastern South America, consistent with patterns observed during ENSO events [120], though it does not reflect the typical rainfall deficits in northeastern Brazil associated to the PSA mode (Figure B.1 B, D). Mode 7 (Figure B.2) reveals a distinct variability pattern in the eastern Indian Ocean, with precipitation teleconnections extending to central Africa and the southeastern United States. This suggests interconnected weather systems across these regions, highlighting the global reach of this mode. Mode 8 (Figure B.3) is characterized by SST variability primarily north of the Arctic Circle. This mode shows a positive trend combined with slow-frequency fluctuations, indicating a possible link to global warming. Its unique spatial focus may be related to Arctic amplification, where the Arctic region warms faster than the rest of the planet, representing a different, potentially nonlinear aspect of climate variability.

Maximizing Correlation – Applying the CPCCA Framework While the Hilbert MCA framework effectively identifies shared variability between SST and continental precipitation by maximizing squared covariance, there are scenarios where maximizing correlation is equally important, especially when the datasets operate on different scales. Canonical Correlation Analysis (CCA) serves this purpose by finding shared patterns that maximize correlation. However, a common drawback of CCA is that its leading modes may exhibit strong correlations but account for only a small portion of the overall variance in each dataset.

To address this, CCA is often performed on a reduced principal component space, focusing on the most significant modes of variability and thereby reducing computational demands [158, pp. 320-321]. Despite its benefits, this approach can miss important correlations involving lower-variance PCs, as highlighted by Jolliffe [85, p. 223].



content/chapter4/figs/raster/tele_psd.png

To bridge the gap between MCA and CCA, Swenson [160] introduced the Continuum Power CCA (CPCCA) framework. CPCCA partially whitens the datasets using a fractional whitening parameter, α_i , allowing for a continuous adjustment between maximizing correlation (CCA) and squared covariance (MCA). In our study, we determined α_i values of 0.0011 and 0.0013 for the SST and precipitation datasets, respectively, following the objective parameter choice proposed by Swenson [160]. Following our previous approach in this study, we extend the CPCCA framework by using Hilbert complexified in combination with a varimax rotation. Based on the mode stability χ the first 28 modes are rotated. All modes are presented in the appendix (Figure B.14–B.33).

Our analysis reveals that the complexified and rotated CPCCA modes are generally similar to MCA modes but exhibit higher correlations, reflecting the framework's emphasis on correlation over covariance. Consequently, CPCCA modes tend to explain less variance within each dataset compared to MCA modes. Additionally, some mode mixing was observed, such as the combination of MCA modes 4 and 9 into a single CPCCA mode (Figure B.17), indicating sensitivity to the chosen α values.

A particularly noteworthy finding is CPCCA mode 5 (Figure B.18), which displays a unique teleconnection pattern linking SST in the western tropical Pacific to precipitation in northern Europe—a pattern not identified by MCA. This aligns with recent research by Straaten *et al.* [159], who used explainable neural networks to identify this SST pattern as a predictor for European summer temperatures on sub-seasonal time scales. Our results suggest that this source of predictability may also extend to precipitation on seasonal time scales, highlighting an exciting avenue for future research.

4.3 Conclusions

In this chapter, we investigated the shared variability between SST and continental precipitation to better understand atmospheric teleconnections—patterns that connect climate variations across distant regions. By applying varimax-rotated Hilbert MCA to an extended dataset of SST and precipitation, we decomposed the complex, high-dimensional covariability into a few dominant modes of shared variability.

Our analysis successfully identified key modes corresponding to well-established climate phenomena such as the seasonal cycle, ENSO, PDO, AMM, and IOD. These modes align well with known climate indices and teleconnection patterns, reinforcing the validity of our approach. Notably, the use of the Hilbert transform and varimax rotation allowed us to capture both the seasonal cycle and lagged teleconnections within a single framework, without the need to remove the seasonal cycle explicitly. This comprehensive approach provides deeper insights into how SST and continental precipitation are interconnected across the globe.

In addition to these established modes, we uncovered additional patterns of shared variability—specifically modes 5, 7, and 8—that offer new avenues for research. For instance, Mode 5 resembles the Pacific-South American (PSA) pattern, influencing climate variability in the South Pacific subtropics and potentially offering predictability linked to tropical regions. Mode 7 highlights a distinct variability in the eastern Indian Ocean with teleconnections to central Africa and the southeastern United States. Mode 8 is characterized by SST changes north of the Arctic Circle, suggesting a link to Arctic amplification and offering insights into the regional impacts of global warming.

To explore the shared variability from a different perspective, we extended our analysis by incorporating the CPCCA framework. By combining CPCCA with Hilbert complexification and varimax rotation, we focused on maximizing correlation rather than covariance. This approach revealed modes with higher correlations but lower explained variance, providing a complementary view of the interactions between SST and precipitation. Notably, CPCCA Mode 5 identified a unique teleconnection pattern linking SST anomalies in the western tropical Pacific to precipitation in northern Europe—a pattern not captured by MCA but recently highlighted in other studies as a source of predictability for European summer climate.

Looking forward, applying non-linear methods could enhance our understanding of shared variability, especially for capturing the inherent nonlinearity of phenomena like ENSO. Techniques such as non-parametric kernel methods [1] or neural network approximations [4, 76] offer promising avenues. The recent availability of tools like the Python library `cca-zoo` [23] makes these advanced methods more accessible. However, challenges remain, particularly in interpreting the underlying spatial patterns when using kernel methods due to the pre-image problem. Recent suggestions using linear approximations [17] or composite patterns [61] to overcome this, could make non-linear methods more practical for climate research.

In summary, our study demonstrates that combining varimax-rotated Hilbert extensions with MCA and CPCCA provides a robust framework for identifying and interpreting dominant modes of shared variability between SST and continental precipitation. These methods enhance our understanding of atmospheric teleconnections and offer valuable insights that could improve climate predictions on seasonal to decadal timescales.

Case Study 2: Representation of the Madden-Julian Oscillation

sub-seasonal time scales;

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Proofs

A.1 Squared Covariance is not Preserved Under Rotation

What we want to verify is whether the total squared covariance of the first rotated r modes of $\mathbf{X}$ and $\mathbf{Y}$ is equal to the total squared covariance of the original first r modes. We will see that this is not the case. To prove this, recall that the total squared covariance of $\mathbf{X}$ and $\mathbf{Y}$ is given by the sum of the squared singular values of the cross-covariance matrix $\mathbf{C} = \mathbf{X}^H \mathbf{Y} / (N - 1)$, as shown in Equation (2.36):

$$\|\mathbf{C}\|_F^2 = \sum_{k=1}^M \sigma_k^2. \quad (\text{A.1})$$

Also, recall that the covariance of the rotated expansion coefficients $\mathbf{P}_{\mathbf{R},\mathbf{x},k}$ and $\mathbf{P}_{\mathbf{R},\mathbf{y},k}$ is a diagonal matrix given by:

$$\frac{1}{N-1} \mathbf{P}_{\mathbf{R},\mathbf{x}}^H \mathbf{P}_{\mathbf{R},\mathbf{y}} \stackrel{(2.47)}{=} \mathbf{W}_x \mathbf{W}_y. \quad (\text{A.2})$$

Our first step involves establishing the relationship between the $\mathbf{W}_x$ and $\mathbf{W}_y$ matrices and the original singular values σ_k of the cross-covariance matrix $\mathbf{C}$. We can do this by considering the normalization condition of the k -th column (mode) of the rotated pattern vectors of the variable $s \in [x, y]$:

$$1 = \|\mathbf{V}_{\mathbf{R},s,k}\|^2 \quad (\text{A.3})$$

$$= \mathbf{V}_{\mathbf{R},s,k}^H \mathbf{V}_{\mathbf{R},s,k} \quad (\text{A.4})$$

$$\stackrel{(2.45)}{=} \left(\mathbf{V}_s^H \Sigma^{1/2} \mathbf{R}_{:,k} (\mathbf{W}_s^{-1})_{kk} \right)^H \left(\mathbf{V}_s^H \Sigma^{1/2} \mathbf{R}_{:,k} (\mathbf{W}_s^{-1})_{kk} \right) \quad (\text{A.5})$$

$$= (\mathbf{W}_s^{-1})_{kk}^2 \left(\mathbf{R}_{:,k}^H \Sigma \mathbf{R}_{:,k} \right) \quad (\text{A.6})$$

Solving for $(\mathbf{W}_s)_{kk}$, we find our desired expression:

$$(\mathbf{W}_s)_{kk}^2 = (\mathbf{R}_{:,k}^H \Sigma \mathbf{R}_{:,k}) \implies \mathbf{W}_x = \mathbf{W}_y \equiv \mathbf{W}. \quad (\text{A.7})$$

Note that this also implies that the normalizing diagonal matrices $\mathbf{W}_s$ are the same for both variables x and y , so that the covariance is $\mathbf{W}^2$.

So, the total squared covariance of the first r rotated modes is then given by,

$$\begin{aligned} \text{tr } \mathbf{W}^4 &= \sum_{k=1}^r ((\mathbf{R}^H \Sigma \mathbf{R})_{kk})^2 \\ &\leq \text{tr} \left((\mathbf{R}^H \Sigma \mathbf{R})^2 \right) \\ &= \text{tr} (\Sigma^2) \\ &= \sum_{k=1}^r \sigma_k^2, \end{aligned}$$

which shows that is less than sum of the squared singular values of the cross-covariance matrix, i.e. the total squared covariance before rotation. This implies that the total squared covariance is not preserved under rotation, except for the trivial case where the rotation matrix is the identity matrix.

In contrast, however, we can show that the total covariance *is* preserved under rotation:

$$\begin{aligned} \text{tr } \mathbf{W}^2 &= \sum_{k=1}^r (\mathbf{R}^H \Sigma \mathbf{R})_{kk} \\ &= \sum_{k=1}^r \left(\sum_{i=1}^r \mathbf{R}_{ik}^H \Sigma_{ii} \mathbf{R}_{ik} \right) \\ &= \sum_{k=1}^r \left(\sum_{i=1}^r \sigma_i \mathbf{R}_{ik}^2 \right) \\ &= \sum_{i=1}^r \sigma_i \left(\sum_k \mathbf{R}_{ik}^2 \right) \\ &= \sum_{i=1}^r \sigma_i. \end{aligned}$$

where we made use of the unitary property of the rotation matrix that $\sum_k \mathbf{R}_{ik}^2 = 1$.

A.2 Frobenius Norm of the Cross-Covariance Matrix

The total squared covariance between $\mathbf{X} \in \mathbb{C}^{N \times P}$ and $\mathbf{Y} \in \mathbb{C}^{N \times Q}$ is quantified using the Frobenius norm $\|\cdot\|_F$ of the cross-covariance matrix $\mathbf{C} = \mathbf{X}^H \mathbf{Y} / (N - 1)$:

$$\|\mathbf{C}\|_F \equiv \left(\sum_i^N \sum_j^P |c_{ij}|^2 \right)^{1/2} = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{C}) \right)^{1/2}. \quad (\text{A.8})$$

For the proof of Equation (A.8), consider the singular value decomposition of $\mathbf{C}$,

$$\mathbf{C} = \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^H, \quad (\text{A.9})$$

where $\mathbf{V}_x \in \mathbb{C}^{P \times P}$ and $\mathbf{V}_y \in \mathbb{C}^{Q \times Q}$ are unitary matrices, and $\mathbf{\Sigma} \in \mathbb{C}^{P \times Q}$ is a diagonal matrix with singular values $\sigma_i(\mathbf{C})$ on the diagonal.

Using the orthonormality of $\mathbf{V}_x$ and $\mathbf{V}_y$ as well as the property that the trace of a matrix remains invariant under cyclic permutations, we find:

$$\begin{aligned} \|\mathbf{C}\|_F^2 &= \|\mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^T\|_F^2 \\ &= \text{tr} \left((\mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^T)^T (\mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^T) \right) \\ &= \text{tr} (\mathbf{V}_y \mathbf{\Sigma} \mathbf{V}_x^T \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^T) \\ &= \text{tr} (\mathbf{V}_y \mathbf{\Sigma}^2 \mathbf{V}_y^T) \\ &= \text{tr} (\mathbf{\Sigma}^2) \\ &= \sum_{r=1}^k \sigma_r^2. \end{aligned}$$

Case Study

B



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Figure B.1 — Caption Description



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Figure B.2 — Caption Description



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Figure B.3 — Caption Description



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Figure B.4 — Caption Description



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Figure B.5 — Caption Description



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Figure B.6 — Caption Description



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Figure B.7 — Caption Description



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Figure B.8 — Caption Description



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Figure B.9 — Caption Description



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Figure B.10 — Caption Description



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Figure B.11 — Caption Description



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Figure B.12 — Caption Description



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Figure B.13 — Caption Description



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Figure B.14 — Caption Description



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Figure B.18 — Caption Description



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Figure B.19 — Caption Description



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Figure B.20 — Caption Description



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Figure B.21 — Caption Description



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Figure B.22 — Caption Description



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Figure B.23 — Caption Description



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Figure B.26 — Caption Description



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Figure B.27 — Caption Description



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Figure B.28 — Caption Description



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Figure B.29 — Caption Description



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Figure B.30 — Caption Description



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Figure B.31 — Caption Description



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Figure B.32 — Caption Description



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Figure B.33 — Caption Description